



LIBS++

Software for Spectral Analysis

USER MANUAL

Credits:

Applied Laser Spectroscopy Laboratory, ICCOM-CNR, Via Moruzzi 1, 56124 Pisa

Tel: +39 050 315 22 21 / 24

Fax: +39 050 315 22 30

e-mail: vincenzo.palleschi@cnr.it

Marwan Technology Srl

Via del Borghetto 34, 56124 Pisa

Tel: +39 050 3138373

e-mail marwan@df.unipi.it

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Introduction

LIBS++ is a software application devoted to the processing of spectra obtained by the laser-induced breakdown spectroscopy technique, according to the 'Calibration-free' method (CF-LIBS), developed in the LIBS laboratory at IPCF/CNR. The application has been implemented using the programming language C++, and the Borland/Imprise C++ Builder, release 6.0, as the compiler. The application release hereafter described is the 5.0

The complete procedure to be followed for the achievement of quantitative analysis by CF-LIBS method is sketched in section 2 and extensively described in section 3 of this manual. The method relies on the assumption of Local Thermodynamic Equilibrium in the plasma. The correct application of the procedure requires the knowledge of several quantities characteristic of the plasma, such as the electron density and the temperature. However, even in those cases when the complete quantitative analysis cannot be carried out, the singles features available in the software may be exploited independently for a partial processing of the spectrum.

To introduce the new user to the main operations allowed by LIBS++ for spectral processing, a case study is described step by step in section 3, concerning the quantitative analysis of a spectrum collected from a certified aluminum sample. This description is intended for spectroscopists, researchers and students with knowledge of the processes involved in LIBS measurements. Theoretical notions are avoided in this section, while an attempt has been made to provide the essential information needed to use successfully the software. Snapshots of the application windows are diffusely used to this purpose.

1. *Installation and guidelines*

1.1 System requirements

LIBS++ can be installed on IBM-compatible PCs, with Pentium processor or higher, mounting a Windows 9x operating system or higher, with 128 Mbytes RAM or more, and 50 Mbytes hard disk free space or more. The configuration suggested is a Pentium IV/1.5 GHz with 256 Mbytes RAM.

1.2 Software installation

The application can be installed by running the setup.exe file from the installation CD. It is necessary that the application be installed in the default folder (C:\LIBS++).

Raw spectra to be processed can be open from any path in the computer resources; the templates, however, during the processing must be accessed from the working directory (C:\LIBS++\templates).

1.3 Spectral Files format and units

LIBS++ is designed to process LIBS spectra both in the form of ASCII files containing two columns x,y (wavelength, counts) without header and in the Avantes proprietary files (*.roh and *.trt). Tab and space are allowed as column separators; the software cannot open files where columns are separated by commas. Uneven x spacing is accepted by the software.

The wavelength axis is assumed to be expressed in Angstrom; the nanometer unit is also allowed and automatically converted to Angstrom. Allowed spectral range in nanometer is 1÷1499, in Angstrom is 1500÷15000. All over the processing, the unit Angstrom is used by the software and in the databases.

1.4 Spectral range and elements

LIBS++ is suited to process spectra covering the near UV – visible - near IR spectral range. The embedded database includes atomic lines (from neutral and first ionization stage) ranging from 2000 to 9000 Angstrom. Spectral data have been collected from NIST DAS and Kurucz databases.

Elements from almost all the periodic table are included in the database. However the information available is not the same for every element. The evaluation of a specific element for quantitative analysis by the CF-

LIBS method is possible only when the list of the expected wavelengths for the allowed transitions is available together with the corresponding energy levels, degeneracies and transition probabilities.

The information available at present for the different elements is resumed in the following table. The color code is explained in the legend.

IA	IIA	IIIA		IVA	VA	VIA	VIIA	VIIIA			IB	IIB	IIIB	IVB	VB	VIB	VIIIB	VIII			
¹ H																					² He
³ Li	⁴ Be												⁵ B	⁶ C	⁷ N	⁸ O	⁹ F	¹⁰ Ne			
¹¹ Na	¹² Mg												¹³ Al	¹⁴ Si	¹⁵ P	¹⁶ S	¹⁷ Cl	¹⁸ Ar			
¹⁹ K	²⁰ Ca	²¹ Sc		²² Ti	²³ V	²⁴ Cr	²⁵ Mn	²⁶ Fe	²⁷ Co	²⁸ Ni	²⁹ Cu	³⁰ Zn	³¹ Ga	³² Ge	³³ As	³⁴ Se	³⁵ Br	³⁶ Kr			
³⁷ Rb	³⁸ Sr	³⁹ Y		⁴⁰ Zr	⁴¹ Nb	⁴² Mo	⁴³ Tc	⁴⁴ Ru	⁴⁵ Rh	⁴⁶ Pd	⁴⁷ Ag	⁴⁸ Cd	⁴⁹ In	⁵⁰ Sn	⁵¹ Sb	⁵² Te	⁵³ I	⁵⁴ Xe			
⁵⁵ Cs	⁵⁶ Ba	⁵⁷ La	*	⁷² Hf	⁷³ Ta	⁷⁴ W	⁷⁵ Re	⁷⁶ Os	⁷⁷ Ir	⁷⁸ Pt	⁷⁹ Au	⁸⁰ Hg	⁸¹ Tl	⁸² Pb	⁸³ Bi	⁸⁴ Po	⁸⁵ At	⁸⁶ Rn			
⁸⁷ Fr	⁸⁸ Ra	⁸⁹ Ac	+	¹⁰⁴ Rf																	
* Lanthanides				⁵⁸ Ce	⁵⁹ Pr	⁶⁰ Nd	⁶¹ Pm	⁶² Sm	⁶³ Eu	⁶⁴ Gd	⁶⁵ Tb	⁶⁶ Dy	⁶⁷ Ho	⁶⁸ Er	⁶⁹ Tm	⁷⁰ Yb	⁷¹ Lu				
+ Actinides				⁹⁰ Th	⁹¹ Pa	⁹² U	⁹³ Np	⁹⁴ Pu	⁹⁵ Am	⁹⁶ Cm	⁹⁷ Bk	⁹⁸ Cf	⁹⁹ Es	¹⁰⁰ Fm	¹⁰¹ Md	¹⁰² No	¹⁰³ Lw				

Elements	Information available				Quantitative analysis by CF-LIBS allowed
	Neutral		Singly ionised		
	Transition wavelengths	Spectral parameters	Transition wavelengths	Spectral parameters	
	YES	YES	YES	YES	YES
	YES	YES	YES	-	YES
	YES	YES	-	-	YES
	YES	-	YES	-	-
	YES	-	-	-	-
	-	-	-	-	-

WARNING: the arsenic symbol ("As") is a key word in the programming language and its use is not allowed as a variable. Thus this element in the database has been temporarily renamed as 'Aq'.

Provided additional or more reliable data are available, the databases can be modified by the user in order to cover specific needs.

2. Quantitative analysis by CF-LIBS at a glance

Before describing the practical operations needed to accomplish a quantitative analysis according to the CF-LIBS method, the schematic flow chart of the procedure is here provided. In fact, the procedure requires several actions and decisions to be undertaken by the user. It is important, therefore, that the user will be as much aware of the general design as possible.

Let's start by loading a new spectrum. The following steps will be undertaken in the order:

- 1- Marking the lines observed in the spectrum
- 2- Identifying the emitting species for as many lines as possible
- 3- Fitting of the lines to obtain the integral intensity values
- 4- Evaluation of the electron density n_e from line broadening
- 5- Evaluation of the temperature T by the Saha-Boltzmann plot
- 6- First calculation of the composition via CF-LIBS algorithm, using the values n_e and T
- 7- Evaluation of the self-absorption effect on the different lines, based on the first estimates of the concentration, T and n_e , and optical path length.

❖ *If the self absorption is negligible, the procedure stops here, and the result of the quantitative analysis is given by the composition calculated in step 6. If, however, self absorption is affecting the intensity of the lines, other steps must be accomplished:*

- 8- Correction of the points in Boltzmann and Saha-Boltzmann plots
- 9- New evaluation of the temperature, based on the corrected points in the Saha-Boltzmann plot
- 10- New calculation of the composition via CF-LIBS algorithm, using the values n_e and T' and the corrected points in the Boltzmann plot

❖ *The procedure in principle stops here. However, it may be advisable, especially when large differences are observed between the compositions obtained in step 6 and 10, to check whether the new scenario does change substantially the evaluation of the self absorption. To this purpose:*

- 7'- Evaluation of the self-absorption effect on the different lines, based on the latest estimates of the concentration, T and n_e , and optical path length.
- 8'- Correction of the points in Boltzmann and Saha-Boltzmann plots
- 9'- New evaluation of the temperature, based on the lately corrected points in the Saha-Boltzmann plot

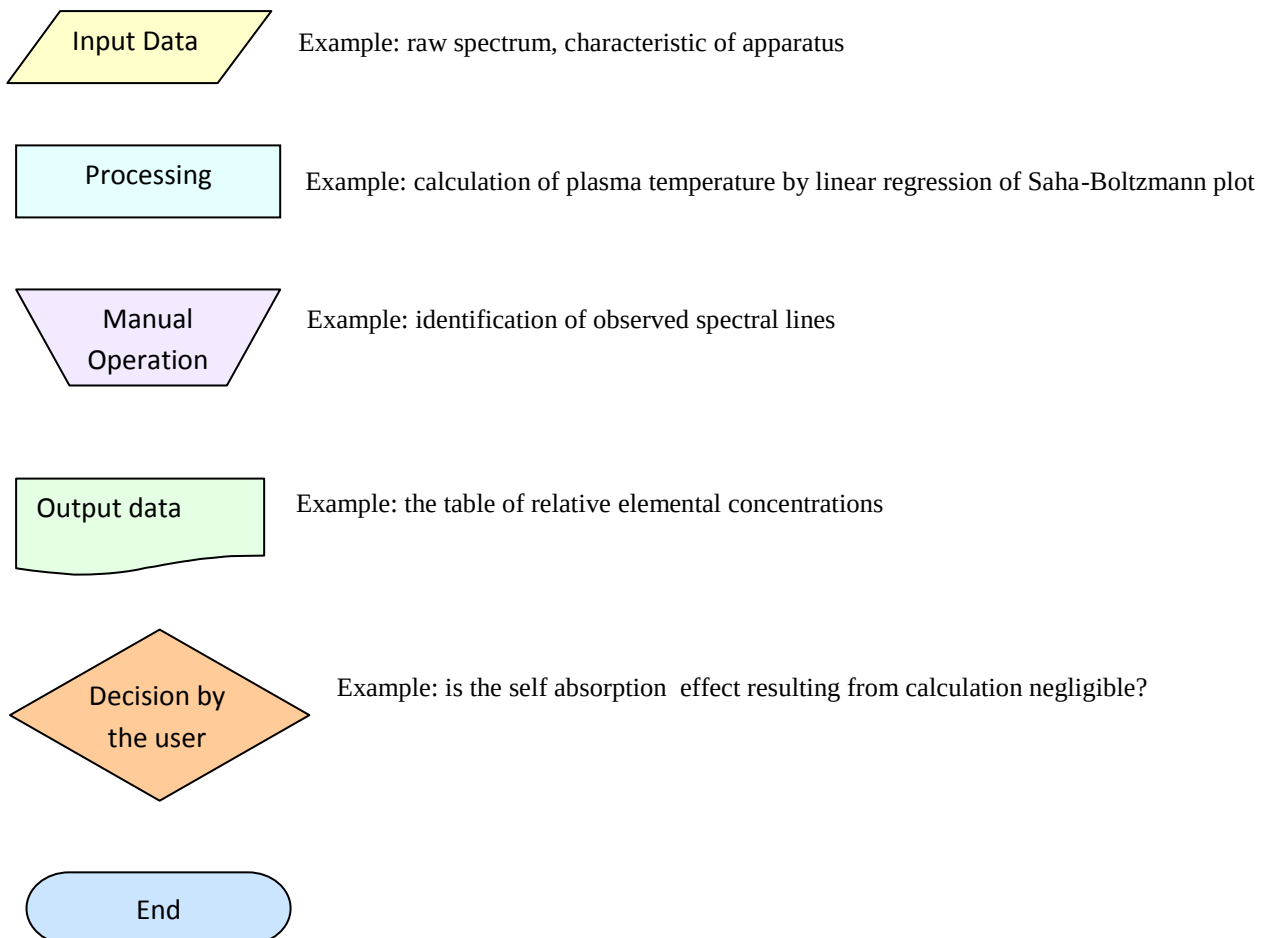
10'- New calculation of the composition via CF-LIBS algorithm, using the latest values n_e and T and the corrected points in the Boltzmann plot

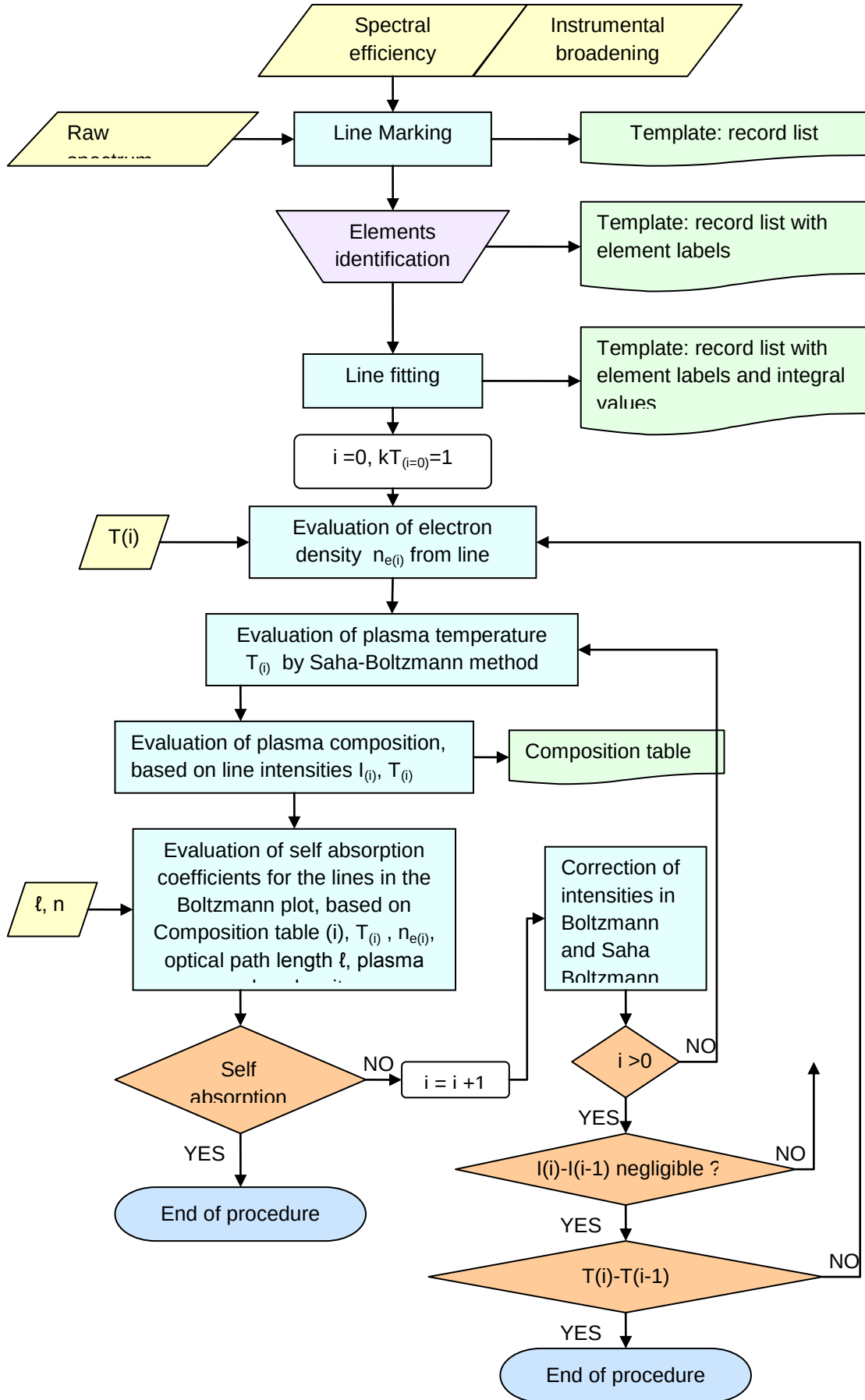
Steps 7'-10' can be repeated until convergence of the results (basically, the evaluation of the temperature should converge within 1%).

Concerning the evaluation of the electron density, in general a refinement by iteration will not be necessary, especially if the hydrogen alpha line is used for the calculations. In the uncommon case that the temperature of the plasma is very far from the default value of 1 eV used for electron density evaluation, the user may want to refine this value too. To this purpose, after obtaining convergence of the temperature value, run again the routine designed to calculate n_e from Stark broadening, after setting the proper value of T . If the new value of the electron density is within 5-10 % of the previous one, no action is required. If a larger difference is obtained, then steps 6 to 10 should be repeated using the latest values of T and N_e .

In the flow chart shown below, it is assumed that the characteristic response of the apparatus is known, in the form of the spectral efficiency and instrumental broadening curves.

Meaning of symbols is as follows:





3. *LIBS++ for Beginners*

In this section a case study will be undertaken and the reader will be guided step by step through the analysis of a certified aluminium alloy sample. This will offer the occasion to illustrate the main features of the LIBS++ software. Snapshots of the working windows are used to this purpose.

3.0 Necessary input data

The LIBS++ software is designed to implement the CF-LIBS procedure, whose main characteristic is the capability to provide quantitative analysis of unknown samples, without need of calibration curves. However, several ingredients are still necessary for the quantitative analysis of a LIBS spectrum by the CF-LIBS method.

3.0.1 The spectrum

First of all, a good quality broad-band spectrum is necessary. In fact, the CF-LIBS procedure allows the evaluation of the sample composition by exploiting the intensity of the lines emitted by all the elements in the spectral region available. Elements whose even most intense lines are undetected (because too weak, or out of the measured spectral range, or masked by overlapped lines from other elements) will not be included in the calculated composition. It is evident, therefore, that high spectral resolution is desirable in order to decrease the effect of spectral interference between nearby lines. On the other hand, a broad band spectrum is desirable in order to determine the composition of unknown samples. Also, the choice of the temporal settings of acquisition is decisive to achieve a good signal-to-noise ratio, which is of course beneficial for the detection of minor elements.

3.0.2 The spectral efficiency of the detection system

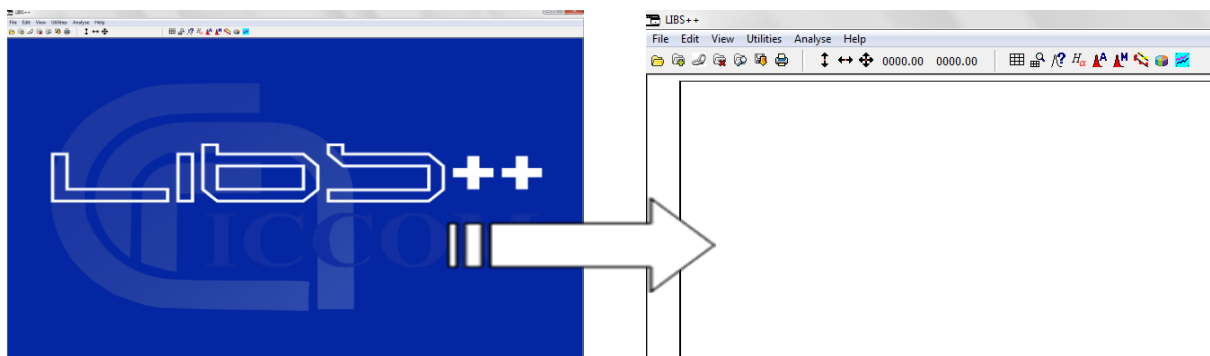
Second, the spectral efficiency of the detection apparatus must be known over the spectral range included in the acquisition. Differently from the analysis based on the calibration curves, here the intensity values of different lines must be compared, even if originated in spectral regions very far from each other. Since the spectral efficiency of the grating, of the sensor and other optical components is not flat in wavelength, the intensity of lines emitted at different wavelengths must be normalised before being used in the CF-LIBS algorithm.

To this purpose, the spectral efficiency of the specific equipment used to acquire the spectrum should be measured in the whole spectral range of interest. The measured behavior should be formatted and stored as a two columns ASCII file. The use of this file will be described later on.

Otherwise, in the case the spectral efficiency of the spectrometer is not known, an alternative method can be used, provided the samples to be analyzed have a similar composition and that the composition of at least one of them is exactly known. The procedure to be adopted in this case, called One Point Calibration (OPC), will be described at the end of this chapter (sect. 3.8.1).

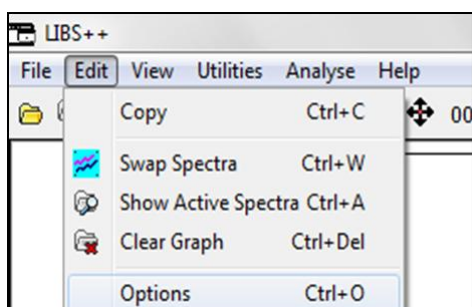
3.1 First operation

When launched, LIBS++ displays shortly a splash screen and then the working window with an empty frame to accommodate the spectrum is shown.

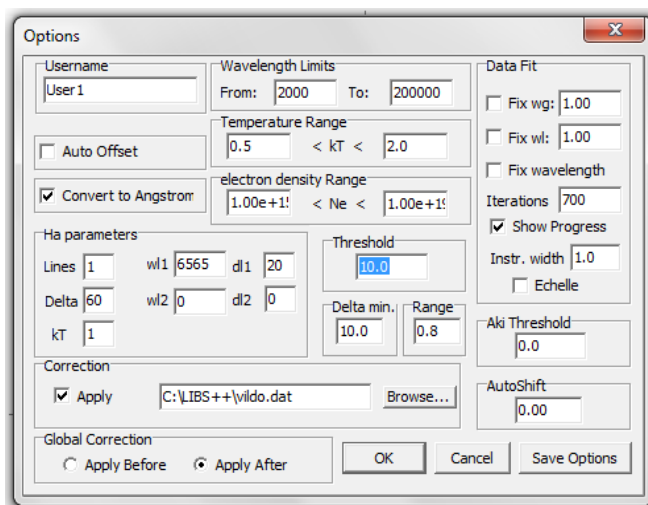


This will be referred in the following as the main window in LIBS++.

Before loading a spectrum, let's take a look to the Options window, which can be displayed by the menu command *Edit/Options*.



This window resumes the current settings applied to every spectrum to be loaded.



LIBS++ allows the user to customize the settings according to the personal preferences or needs. After setting properly the different parameters, the configuration can be saved by the "Save Options" button.

Moreover, the Options settings are saved automatically when the LIBS++ application is closed. The new configuration is saved in the file LIBS++.ini in the working folder (C:\LIBS++), overwriting the previous configuration.

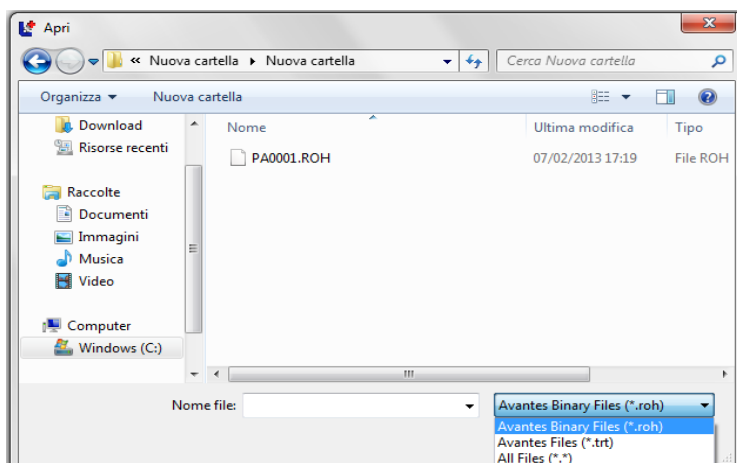
For this first look, it will be sufficient to mention only few of the features included in the Options window, while the other will be described at the proper time.

It is worth to point out that the spectral range of interest can be set by typing the desired lower and higher limits in the “Wavelength Limits” boxes, in Angstrom units. The default spectral range is $2000 \div 9000$ Angstrom. When the spectrum loaded is shorter, the spectral range spanned in the main window is automatically rescaled to the extremes of the data available.

On the other hand, when the spectrum is wider than the limits set here, only the “allowed” range is loaded and displayed. To display the data corresponding to spectral regions external to the limits, just change the limits setting and load again the file.

3.2 Loading a spectrum

The main operations on the spectra can be performed by means of the icon buttons available on the upper left side of the main window. First of all, let's load the spectrum to be processed by means of the  “OPEN” button .



The folder displayed by default to select the file is the latest folder accessed by LIBS++.

The default extension for spectral files is .roh, i.e., Avantes binary files; however, Avantes files *.trt as well as files with any other legal extension may be accessed, provided they are, the latter, ASCII files composed of two columns (wavelength, counts).

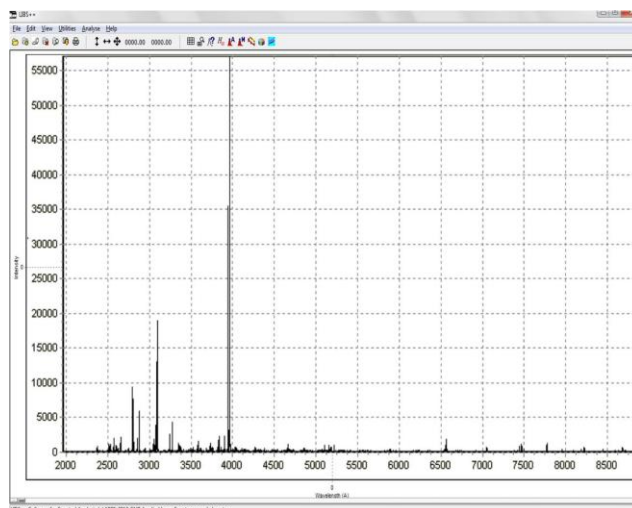
The list of the files with generic extension can be obtained by selecting “All files (*.*)” in the file type box.

The file loaded is displayed in the main window as shown below. The name of the file is quoted on the edge of the window. Wavelength unit is Angstrom. The intensity values on the Y axis are as listed in the file.

The spectrum chosen for this case study has been acquired by an echelle spectrometer coupled to an ICCD, after a double pulse, characterized by two 40 mJ, 1064 nm, 10 ns width collinear laser pulses, separated by 2 μ s.

The focusing lens had a focal length of 100 mm. The distance lens to sample was about 95 mm.

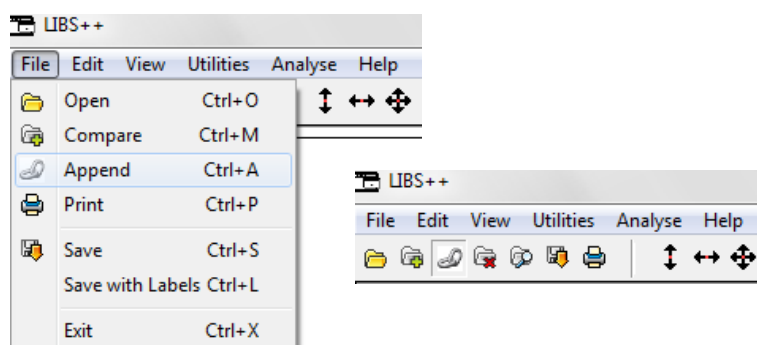
Acquisition settings were: delay after second pulse 3 μ s, gate 1 μ s. The spectrum is the result of accumulation of 10 double shots on the same spot. The emission signal, collected by a fiber optic, was spatially integrate.



3.2.1 Merging two spectra

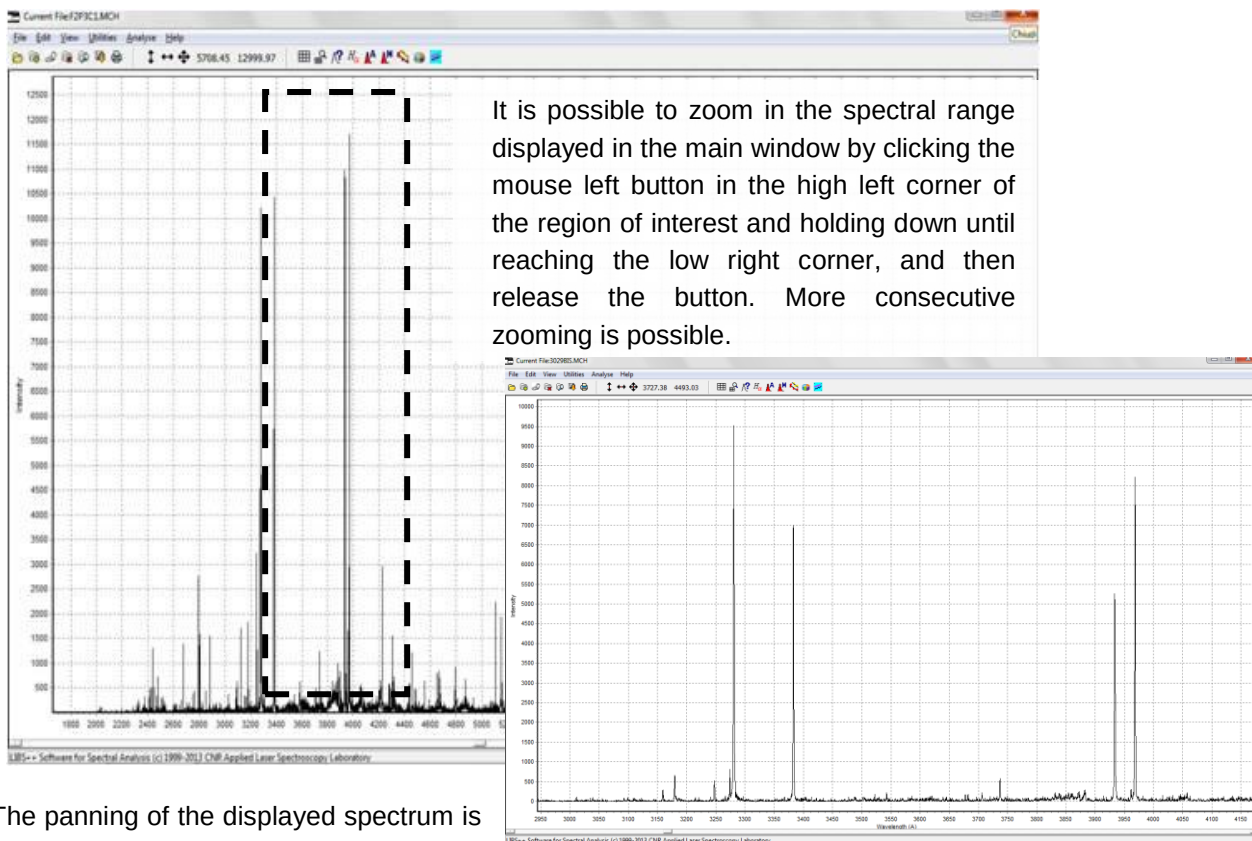
In some cases, spectra at different spectral ranges are acquired simultaneously. The Modi 'Smart' is equipped with an Avaspec spectrometer consisting of two gratings which allow the simultaneous acquisition of two LIBS spectra from 180 to 430 nm and from 380 to 900 nm. The two spectra can then be merged in a single one.

a) Let's open the first spectrum, that is the one at lower wavelength, as described in Sec. 3.2. Then, press on the "Append" button, that is the third from the left on the toolbar. A new window will open, in which all the spectra are listed; at this point, choose the one to append and click on it: in the LIBS window the new spectrum will be appended to the first one and they will appear as a single spectrum. In the superposition region, the resulting spectrum will be the average of the two original ones. Note that the order in which the spectra are opened is mandatory: the first spectrum MUST be the one at lower wavelength and the appended one the one at higher wavelength.



b) Instead of opening the first spectrum, let's click on the "Append" button and then select the two complementary files; in this way, the two spectra will be opened in the LIBS++ main window as a single one.

3.3. Exploring the spectrum



The panning of the displayed spectrum is activated by holding down the right button of the mouse. A scroll bar on the lower edge of the main window also allows rapid positioning along the spectral range.

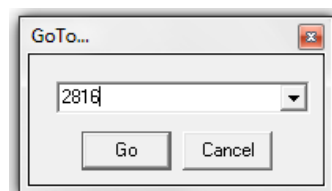
To restore the display of the whole spectrum it is possible to use the menu command *View/Show All*, or the equivalent icon . It is also possible to click and drag the mouse left button in the direction opposite to the *zoom in* operation (toward left-high).

The full scale view can be restored separately for X and Y axes by the commands *View/Auto X* and *View/Auto Y*, respectively (or by using the corresponding icons  and ).

In case the user is interested in a specific spectral region, the menu command *Utilities/GoTo...* may be useful. A dialog box is displayed, allowing to type the central wavelength of interest (in Angstrom).

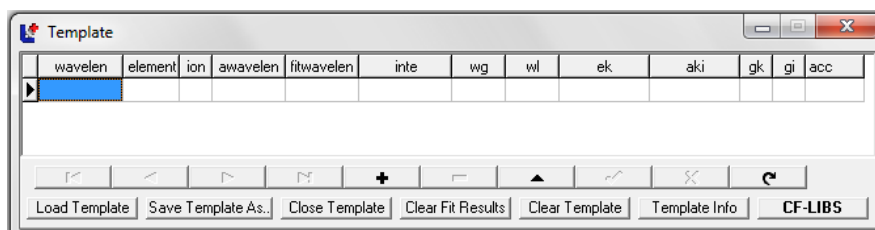


As a result, the view in the main window is zoomed in to show a spectral region centered at the wavelength specified and wide $2 * \text{Delta min}$, where Delta min is one of the parameters set in the Options window.



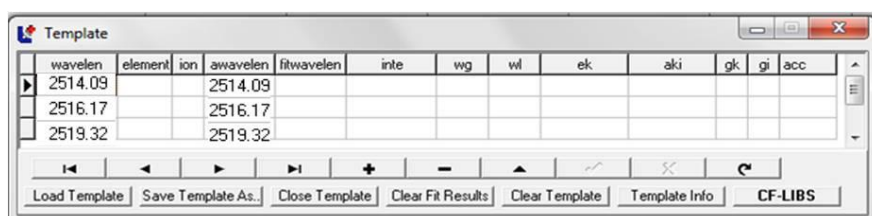
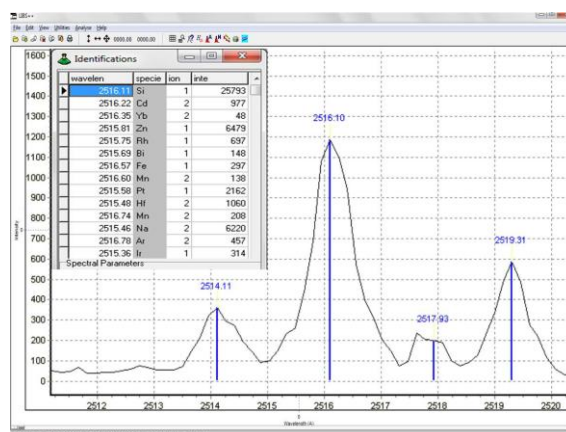
3.4 Marking the peaks

Information useful for qualitative and quantitative analysis is contained in the atomic and ionic emission lines. The first goal in the processing is thus to build a list of the lines detected in the spectrum. This list will be stored in a table called "TEMPLATE", shown in a different window. Each spectral line is listed in the template as a separate record, characterized by several fields, initially empty, to be filled as far as the processing advances, as described in the following.



The list of the lines can be built manually or automatically.

Manual marking of the spectral lines is realized by pointing the mouse to the desired wavelength position (attention should be paid to match the central peak position as closely as possible) and holding down <Shift>+<LeftButton>. The visible effect in the spectrum will be the drawing of a blue segment with blue label indicating the wavelength, while a new record will be added to the template.

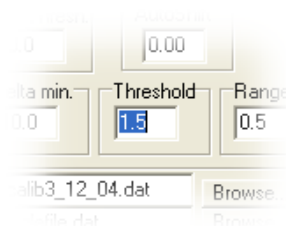


The list of the lines in the template is ordered by ascending wavelength, whatever is the order they have been marked.

Marked lines can be individually deleted by pointing the mouse cursor to the segment position in the main window (the pointer will turn to a hand shape ) and holding down <Shift>+<Rightbutton>. The blue segment will be canceled from the spectrum and the corresponding record deleted from the template.

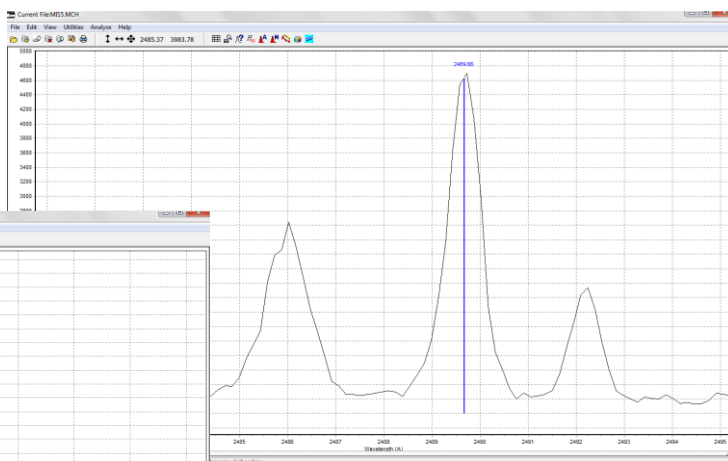
An automatic routine can also be exploited to detect, mark and list in the template the spectral lines appearing in the spectrum: to do this, use the menu command *Analysis\Find Peaks* in the main window or the icon .

This command activates a screening algorithm based on the evaluation of the second derivative of the spectral profile. The value of the second derivative calculated point by point is compared to the currently set threshold in the Options window: if the second derivative locally exceeds the threshold, a line is marked at the corresponding wavelength.



At a fixed value of threshold, the efficiency of the algorithm for the recognition of emission peaks is strongly dependent on the characteristics of the spectrum (spectral resolution, noise level, etc.). On the other hand, the threshold may be tuned by the user to optimize the efficiency, case by case. The default value is 1.5. Recommended values are between 1 and 10.

If the threshold is high (let's say 6), only the sharpest peaks will be recognized and marked.

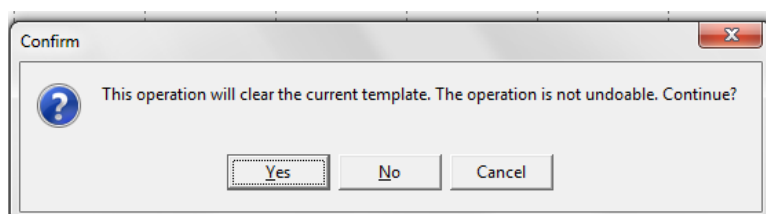


With low threshold values (let's say 1.5), even less prominent features, such as smooth peaks or shoulders, are marked.

The command Find Peaks is active on the whole spectral range of the file loaded, independently on the region displayed in the main window at the moment.

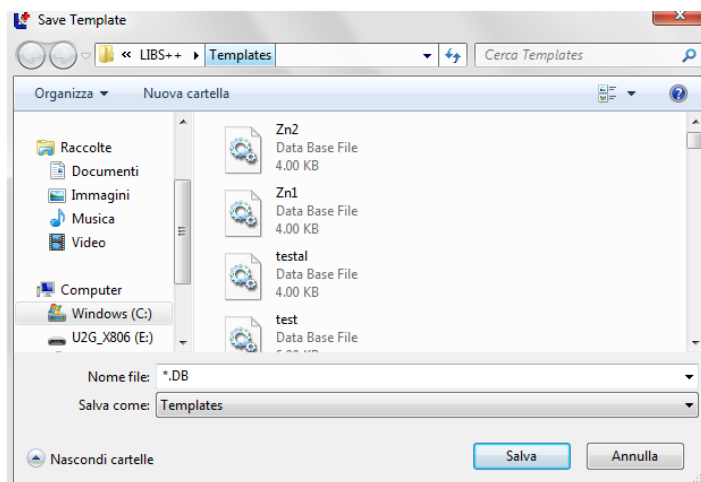
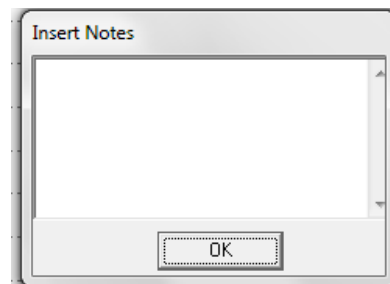
It is important to bear in mind that using the automatic *Find Peaks* routine will clear the existing records in the template and substitute them with the list newly created. At any use of this routine, the user is recalled by a request of confirmation:

Yes confirms the will to execute the find peak routine; No and Cancel return to the working window without executing the routine.



It is recommended to use the automatic peak finding as a first step in the processing, followed by a manual refining (addition or removal of marks). Do not invert this order!

Whatever method has been used to mark the lines, manual or automatic, the list contained in the template should be saved by the command “Save template as” in the template window. When using this command, the user is first allowed to write down some notes, which will be saved in the subsidiary text file created in association to the template. Other information saved in the subsidiary file are the name of the spectrum processed, the name of the operator (current Username in the Options window), the spectral efficiency file used for normalizing the intensity etc.



Then, after pressing OK, the default folder is displayed to allow the saving of the template itself. If the template is to be reload in the following, it is necessary to save it in that folder. The template is saved with a default extension .db. Its format is compatible with Access and Excel. The information included in the file is the same that is displayed in the template window during the LIBS++ operation. The name of the template must begin by an alphabetic character and must not include blank spaces.

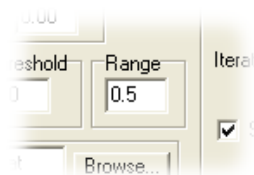
To save our template, let's use the same name of the file to be processed: the new template has now been created.

3.5 Lines identification

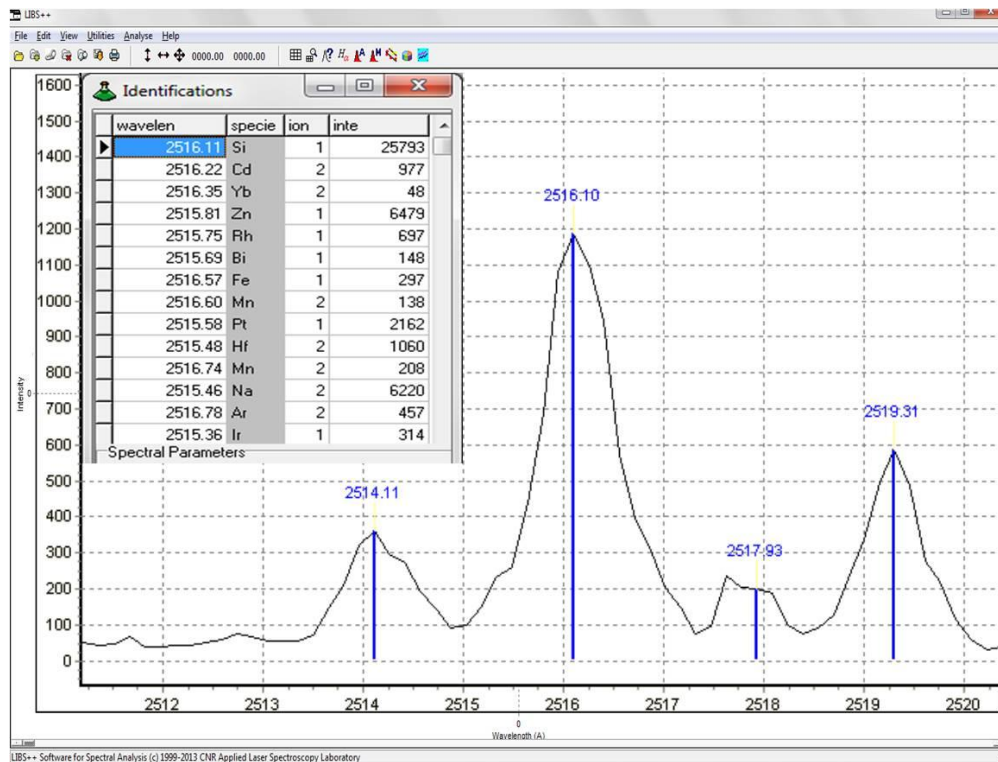
The attribution of the lines recognized in the spectrum to the proper emitting elements is performed by comparison of the observed central wavelength to the records listed in the embedded spectral database (compiled by merging data from NIST and Kurucz files).

To search the database for the possible transitions characterized by a specific wavelength, double click with the left mouse button in the desired wavelength position in the main window. The query thereby generated will extract from the database the items fulfilling the following criterion for the transition wavelength λ :

$$I_0 - \Delta\lambda \leq I \leq I_0 + \Delta\lambda$$
 where λ_0 is the wavelength position chosen by the click and $\Delta\lambda$ is the current value of the parameter Range in the Options window (in Angstrom unit).



The spectral lines selected from the database on the basis of the query are then listed in a new window named “Identifications” .



The lines are listed in ascending order of distance from the central wavelength λ_0 , up to a maximum distance $\Delta\lambda$. The table displayed in the identification window provides the following information for each line:

Central wavelength, Emitting element, Ionization stage, Relative expected Intensity
 (Column: wavelen) (Col: Specie) (Col: ion) (Col: inte)

The ionization stage is quoted in spectroscopic notation (I – Neutral, II – singly ionized). The spectral database includes currently only lines from the neutral and singly ionized stages.

In the database, the relative expected intensities are compiled on different scales for different elements and for different ionization stage of the same element.

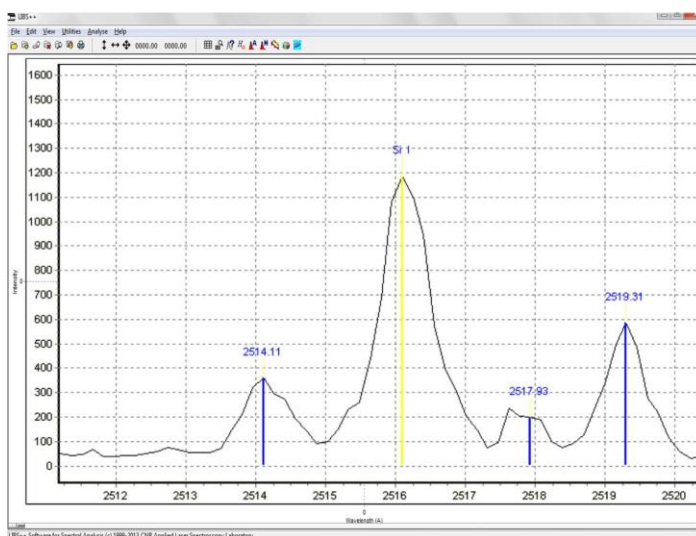
Now, the user is left with the uneasy task to decide which atomic transition among the ones listed (hopefully the list would be complete) is the real source of the observed emission peak. However, this topic is part of a general training in the analysis of LIBS spectra, and is not specific to the CF-LIBS method or to the LIBS++ software. Therefore, let's assume for the moment that the user is able to make a decision; some suggestions about how to proceed in the pattern recognition are given in the next section 3.5.1.

Assume then that the correct identification for the highest experimental peak shown above is the Si I line at 2516.11 Å. To associate this information to the marked line, double click on the row of Si I in the table “Identifications” with the left button of the mouse. This act will copy the spectral information available in the database for the line chosen to the record corresponding to the marked line in the template.

wavelen	element	ion	awavelen	fitwavelen	inte	wg	wl	ek	aki	gk	gi	acc
2510.62			2510.62									
2516.12	Si	1	2516.11					4E04	1.21E00	5	5	0
2519.02			2519.02									

At the same time, the label and the color of the marked line in the main window are changed to indicate the success of the identification.

At least, the information copied in the template includes the expected central wavelength of the line (Column: awavelen, e.g. assigned wavelength), the emitting element (Column: element) and its ionization state (Column: ion).



When available, also the upper level energy (Column: ek) the statistical weights (Columns: upper gk and lower gi, respectively) and the probability of the transition (Column: aki) are filled in the corresponding fields. Energy values are provided in cm^{-1} units, while transition probabilities, in s^{-1} , are divided by 10^8 .

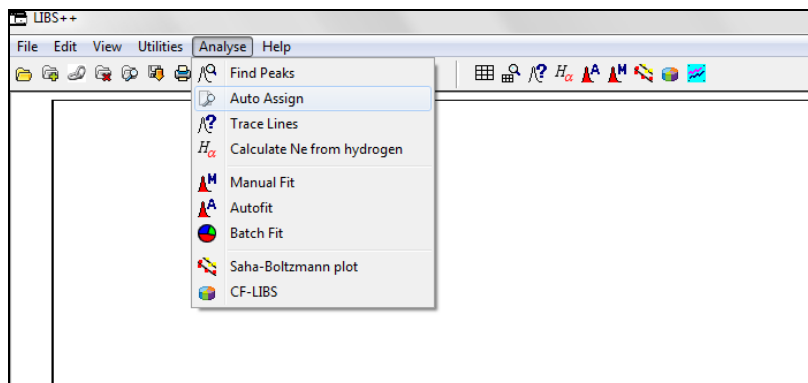
The availability of the spectral parameters for the transition in object is revealed also in the main window by the color of the marked line, which is turned to yellow. In case the spectral information on the transition is not available in the database, the marked line is turned to light blue.

As already mentioned in section 1.4, only lines with known spectral parameters can be used for the quantitative analysis by the CF-LIBS !

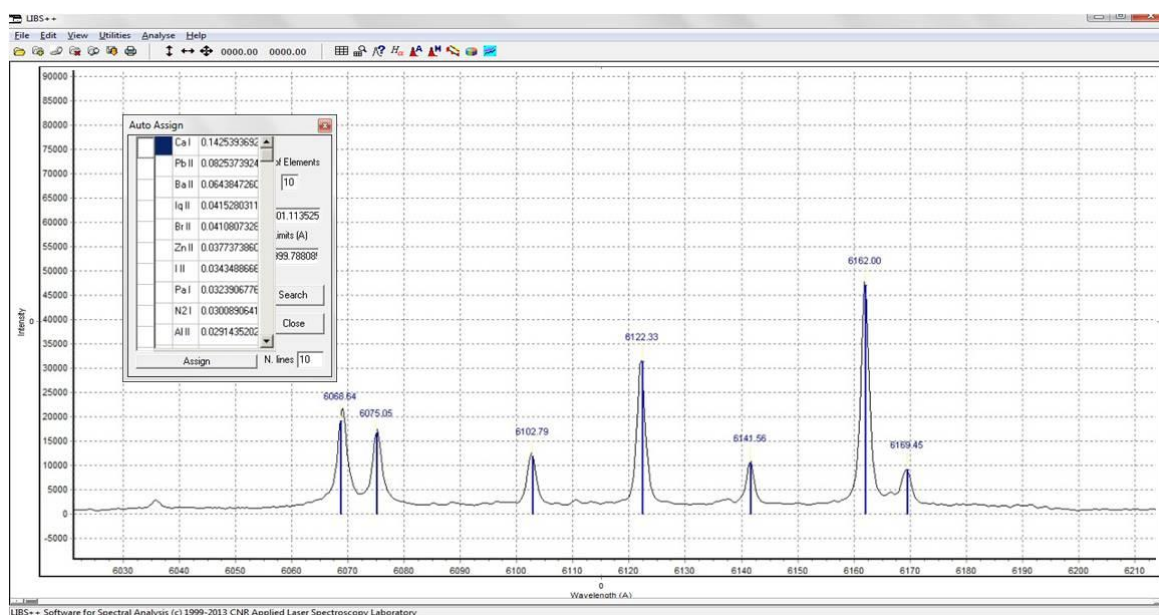
The attribution of each spectral line can be changed independently from the others at any time, just by repeating the steps above described and choosing a new item in the list of the Identifications window.

3.5.1 Unassisted lines identification

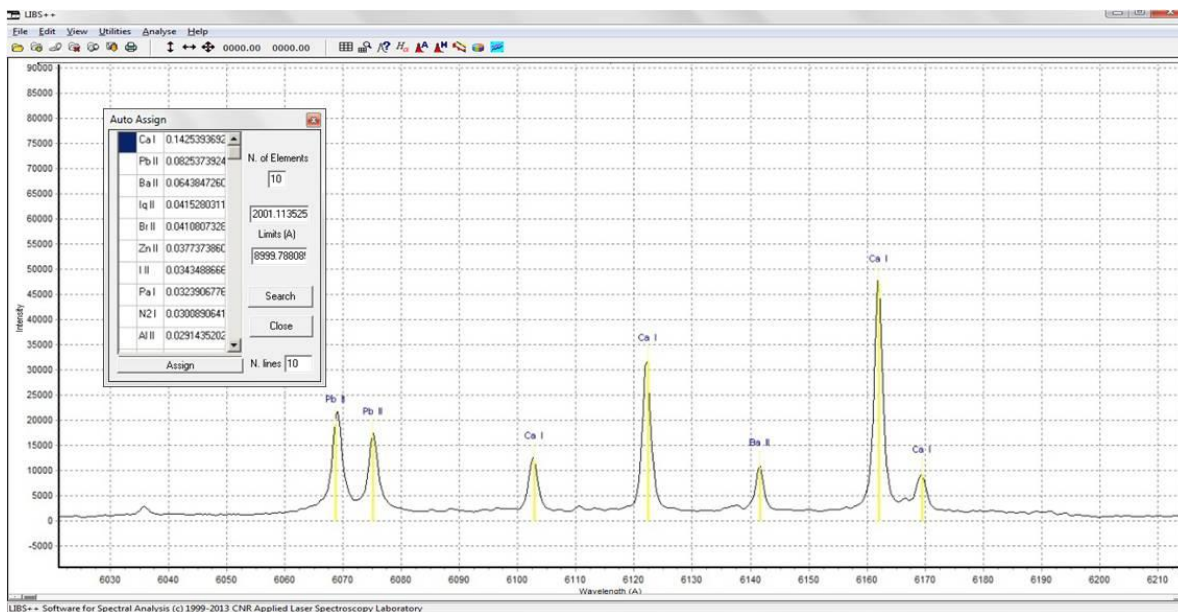
With the latest version of the program, a fully automatic identification of the peaks of the template is available. To perform the unassisted line assignment, go to *Analyse/Auto Assign*.



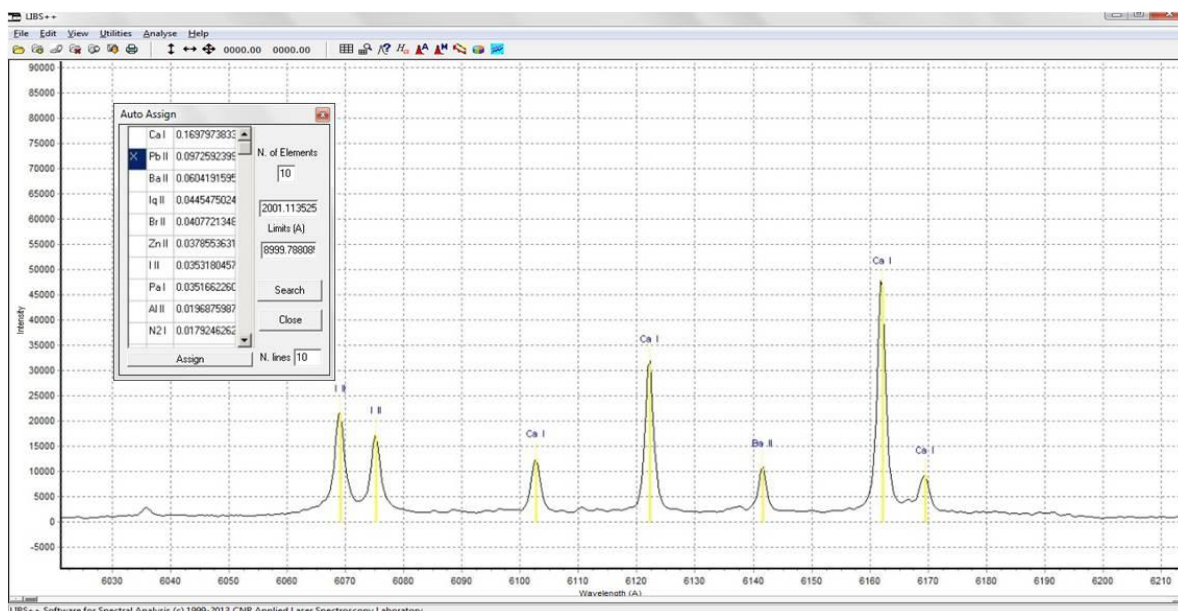
A new window appears with the lower and higher wavelengths limits in which the lines will be assigned. These values are calculated automatically by the program according to the active window.



By clicking on *Search*, a list of elements is calculated by the program and reported on the left column; a score is assigned to each element, providing the best estimate of the probability for the given element to be associated to one of the peaks in the active window. By pressing *Assign*, the program associates each peak to one of the elements, according to its relative score.

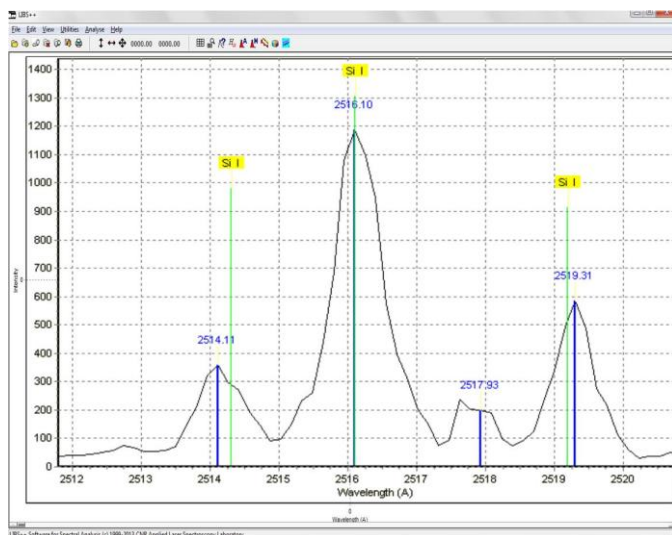
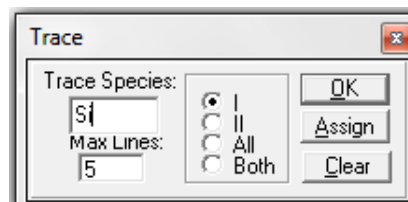


The unassisted assignment is not 100% reliable, and the user should carefully verify its results in order to reject wrong assignment of lines. If the user can be reasonably certain that one of the elements was misidentified by the algorithm, that elements can be excluded from the assignment by clicking on the white box on the left of the species where a cross will appear. Let's suppose that, for example, we are confident that Pb II is not present in the sample; then we just have to click on the corresponding box and to repeat the assignment by clicking Assign. As it can be seen, now the peaks that were assigned to Pb II are now assigned to a different species (I II in this particular case).



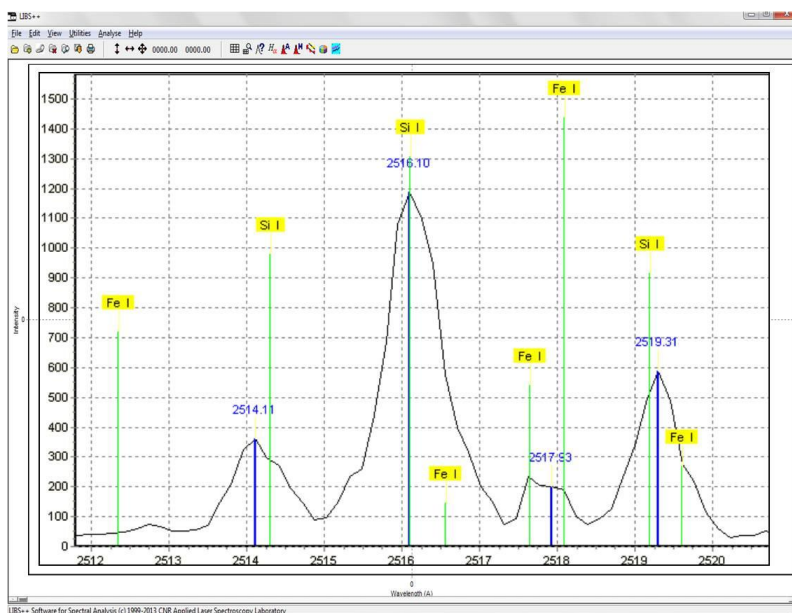
3.5.2 Semi-automatic identification

The command *Analyse/Trace Lines* in the main window is designed to make the identification of many spectral lines easier and more automatic. When using this command, a dialog box is shown, allowing the input of the desired element (in the box "Trace Species"), ionization stage (via the checkbox) and number of lines to be drawn (Max Lines).



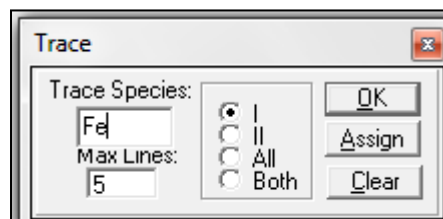
By clicking OK, a query is formulated to extract from the database the n (= Max Lines) most intense lines for the chosen element and ionization stage(s) expected in the spectral range actually displayed in the main window. The result of the query is displayed as a set of n wavelength positions drawn in green color with yellow labels. The expected relative intensity of the traced lines is evidenced by the height of the green segments.

In the specific case, the result obtained from the query about the lines of neutral silicon is shown.



The command Trace can be used repeatedly for different elements or species.

If the traced lines do match the marked spectral lines in wavelength position, the command Assign in the Trace window will automatically copy the spectral information of the traced lines to the records in the template.

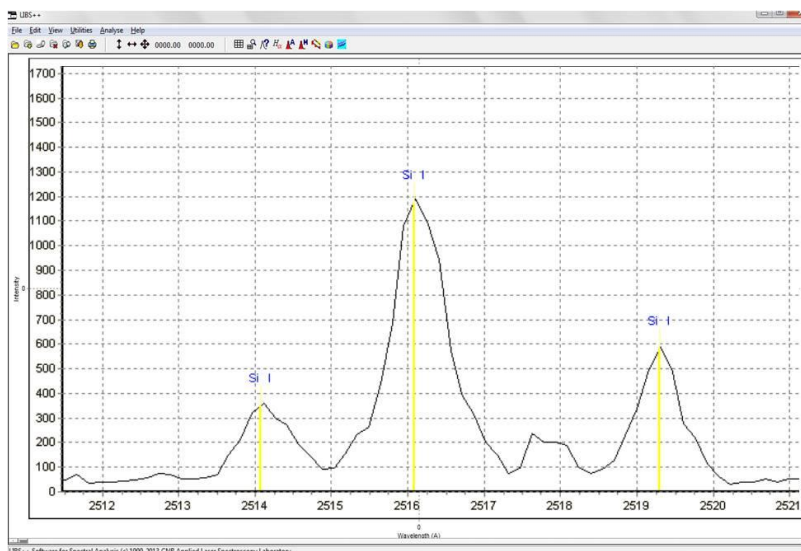




The matching criterion requires that the distance in wavelength between the marked line and one of the traced lines is within the Range parameter value, set in the Options window.

In the specific case, the Assign command will lead to the situation shown here at left.

The matching criterion was fulfilled for the three silicon lines. The iron lines have not been assigned because none marked line was present in their neighborhoods. However, as a general rule, the Assign command applies to all the species already traced in the main window and not only to the last one, which is still displayed in the box.

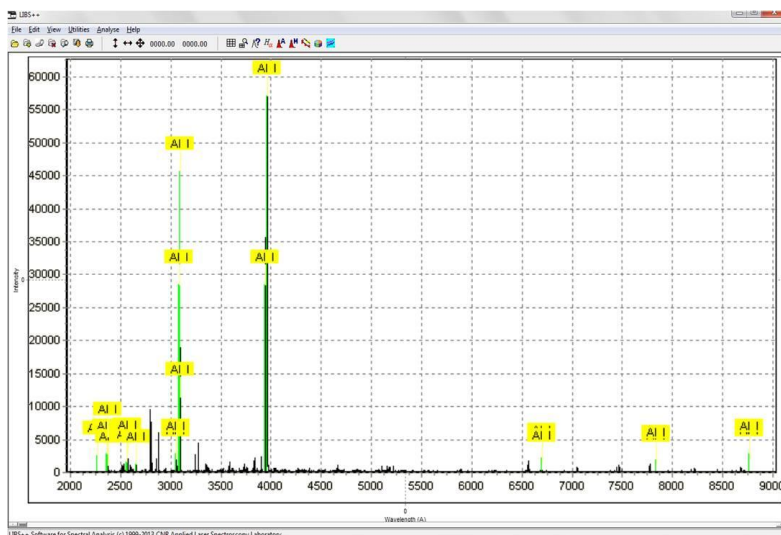


The automatic attribution performed by the Assign command overwrites the earlier attribution, if present !

The user should be aware that this operation is not automatically undoable (anyway single assignments can be manually amended one by one).

After the use of the assignment command, the traced green lines are automatically removed from the main window. If the assignment is not desired, the command Clear in the Dialog box turns off the display of the traced lines.

The availability of the above described tool may be of great help in the pattern recognition of an unknown sample. Of course, it is preferable to have at least a hint about the most abundant element in the sample, but when this is not the case it is however possible to proceed by a trial and error approach. The number of lines to be traced can be adjusted depending on the spectral range available and also on the total number of lines in the spectrum.

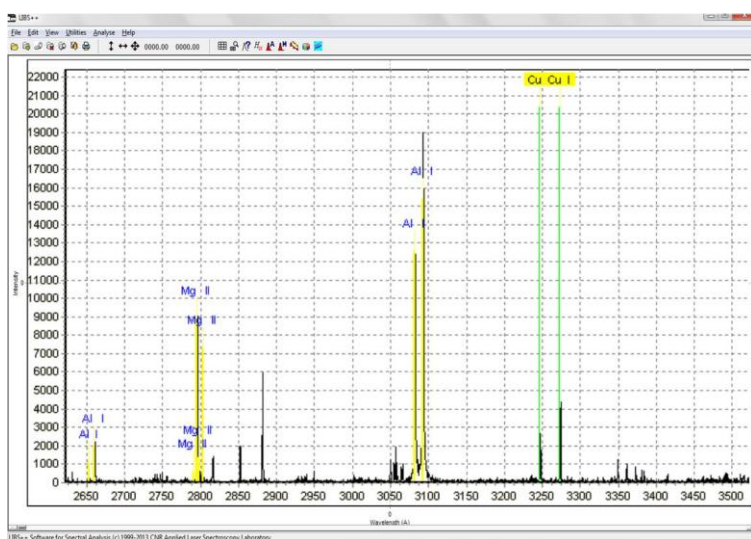
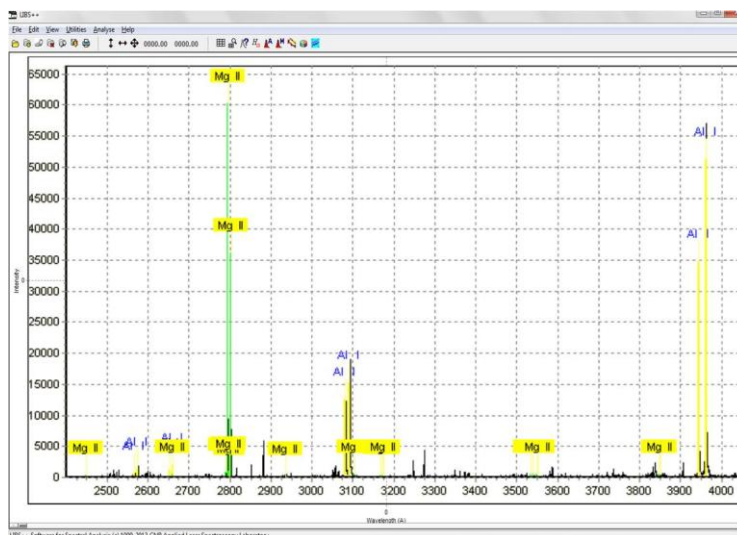


Practically: try first to identify the most intense lines in the spectrum by tracing the pattern of likely elements.

In our case, there is a very good matching of the most intense observed lines with the most intense lines expected from neutral aluminium. Let's assign those lines and then go ahead to consider the next most intense ones.

Now, a very good match is found between the most intense yet unassigned lines and the most intense lines expected from ionised magnesium (of course the assessment of a good match is based on a zoomed view, which is not shown here).

Let's assign also those lines and go to the next.



Again, a matching is observed for the most intense lines expected from neutral copper, so that we can assign also those lines. We can continue with the silicon and so on.. and we should not forget to check the presence of lines from the other ionization state of the elements already identified (we could find then lines from ionised aluminium, neutral magnesium, etc.)

The identification of a specific element will be of course more reliable if based on the matching of more than one line, and

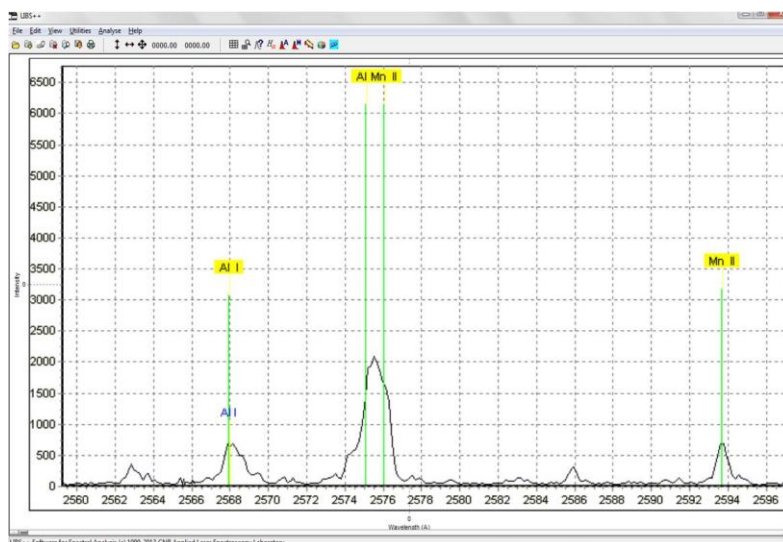
especially if the observed relative intensities correspond to the expected behaviour according to the database.

After the identification of the most intense lines in the spectrum, it may be convenient to switch to a wavelength scan, instead of continuing wondering up and down across the spectral range.

Very likely, some early assignment will appear wrong at a later stage of refinement: however anything can be changed at any time. It may be also advisable to skip the assignment of the uncertain lines: hopefully, the information already gathered may be sufficient to perform the quantitative analysis.

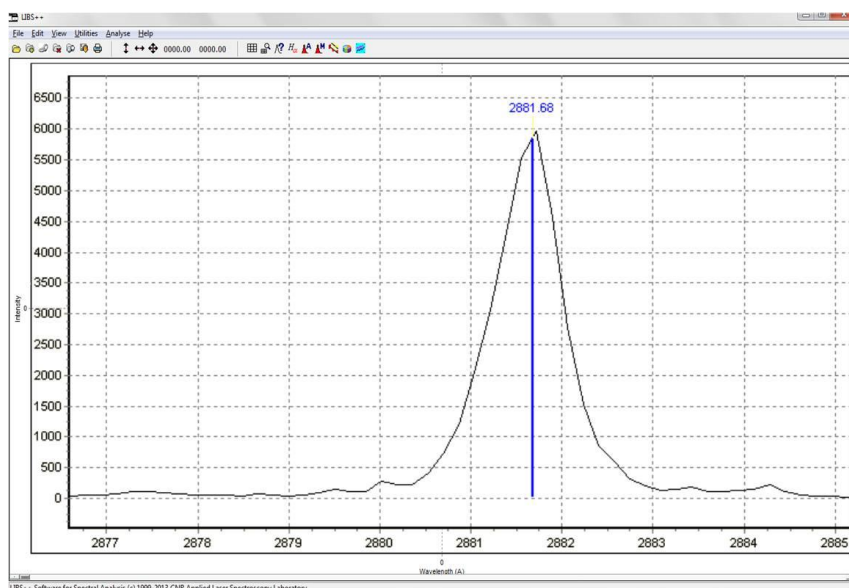
It is strongly recommended to avoid the assignment of lines that are recognised to be the results of the contribution from two transitions (from the same or different elements).

For example, the observed emission peak at 2575 Å is the convolution of a line from neutral aluminium and one from ionised manganese. Attributing the observed peak to any of the two sources would mean to overestimate its intensity, introducing a bias in the following quantitative analysis.



3.6 Peaks fitting

The quantitative analysis of the spectrum requires the determination of the intensity of the lines emitted by the different elements. In particular, the LIBS++ software relies on the evaluation of the integral intensity of the lines. The evaluation of the integral intensity is performed analytically, through the best fitting of the experimental emission profile by an analytical function. In the following a description of the general procedure to be followed for calculating the peak intensities is provided.



Let's start by a good-shaped, isolated line.

First of all, the peak of the experimental profile must be marked as described in section 3.4. In fact, the fitting algorithm will pick from the template the information about the initial estimate of the central wavelength.

Therefore, the corresponding template record must exist before any experimental line can be fitted.

The assignment, on the other hand, doesn't affect the execution of the fitting routine. The assignment of the line to the emitting element can be carried out before or after fitting, indifferently.

Second, the spectral interval displayed in the main window must be adapted to the line to be processed. Avoid tails from other outlier lines, but include a small portion of the baseline.

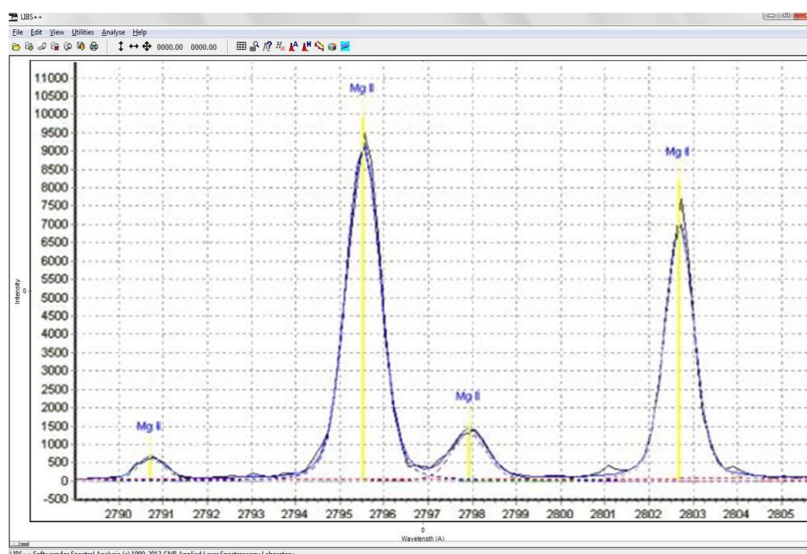
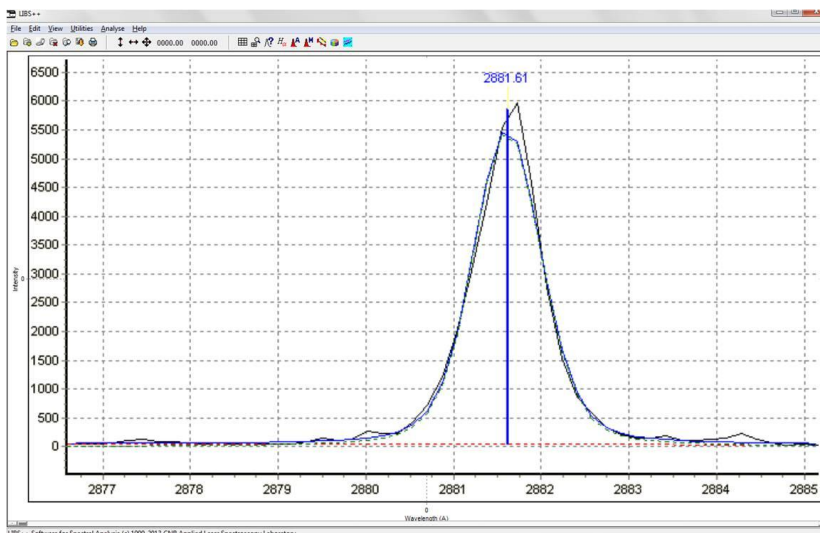
The analytical function used to fit the experimental emission peaks is the Voigt profile. The parameters optimized during the fit are the central wavelength, the height, the width (Gaussian and Lorentzian contributions). The continuum background is also taken in to account by using a linear function with free slope and intercept values.

The optimization of the parameters will be automatically refined until the χ^2 will reach a minimum. At the end of the fitting routine, the integral of the line is analytically computed from the parameters resulting from the fit.

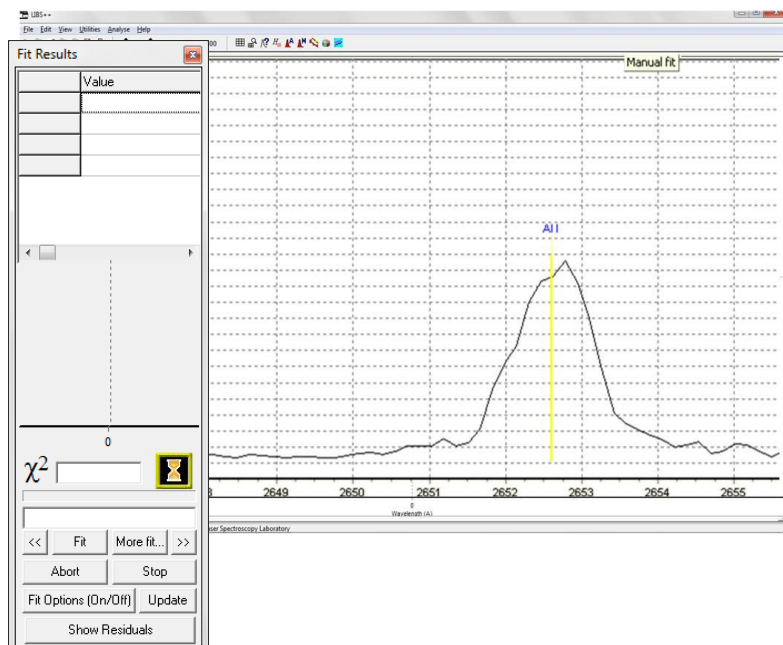
Multifunction fitting is also allowed, to account for overlapped peaks. For example, the multiplet from ionized magnesium can be fitted at once.

However, the fitting of more than 5-6 peaks simultaneously will largely increase the computing time.

In the main window, the analytical profile resulting from the fitting is drawn in blue color (to be distinguished from the experimental profile, in black). The straight line accounting for the background is



colored in red. The individual Voigt functions used in the fit are also drawn for the completeness of information.



Practically, manual or automatic fitting can be chosen by the user.

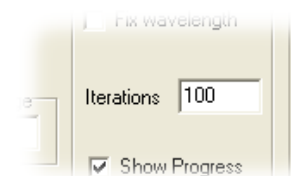
Let's describe first the manual fit mode.

When pressing the Manual Fit icon , the Best fitting routine is directly applied to the spectral range actually displayed in the main window and to all the marked peaks appearing therein.

A new window "Fit Results" is displayed, to allow the user controlling the fitting operations.

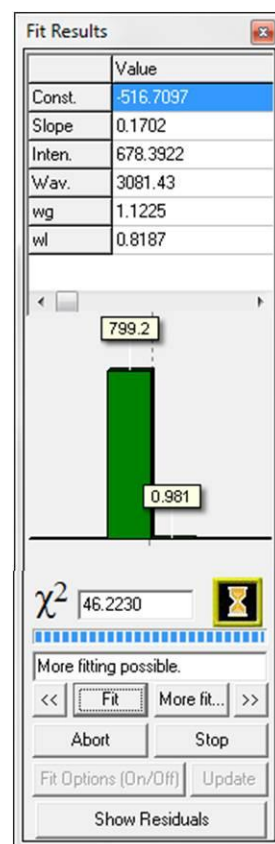
The Fit button in the Fit Results window starts the fitting of the marked lines actually displayed in the main window. Before pressing the Fit button, zooming or panning of the spectral interval included in the main window is still allowed, for example to reduce interferences from the tails of lines outside the displayed range.

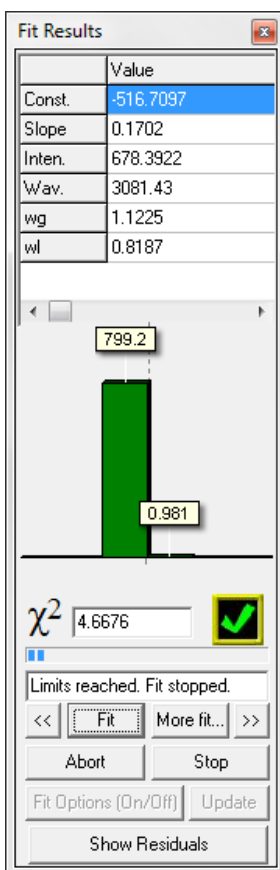
After clicking the Fit button, a first cycle of n iterations is performed automatically,



where n is set by the parameter Iterations in the Options window.

In the same window Fit Results, the values of the fitting parameters after n iterations are provided, listed in separate rows of the upper side table. In the table, intercept and slope of the linear function used to represent the background are first listed, then followed by intensity, central wavelength, Gaussian width and Lorentzian width of the first Voigt function, and, in case, by the same parameters for the other peaks.





Below the table, a histogram offers a graphical perspective on the results. For each line fitted, the histogram shows two bars: the first is related to the integral intensity while the second refers the net width of the line, calculated by unfolding the instrumental broadening from the fitting values.

If the exit criterion (a minimum of the χ^2 parameter) has not been fulfilled yet at the end of the n iterations, a yellow clepsydra in the Fit Results window graphically indicates that a decision is expected from the user.

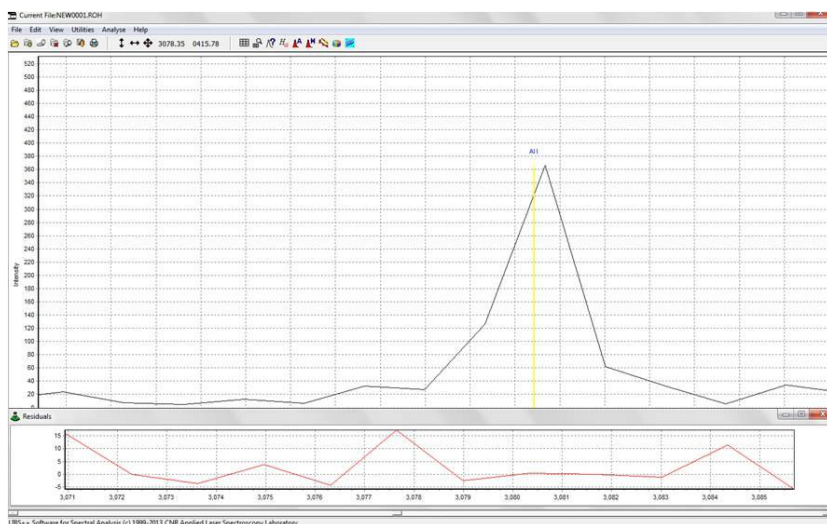
If the user wants to continue with a further fitting cycle of n more iterations, the button "More Fit..." must be used.

If the current result is acceptable, the user should press the button Stop, to exit the fitting routine and copy the results to the template.

Alternatively, to exit the fitting routine and discard the results the Abort command should be used.

When the convergence criterion of the χ^2 is fulfilled within the first cycle of n iterations, the fitting routine is ended and the user is noticed by a green sign.

Also in this case, the decision whether accept the resulting fit (button Stop) or cancel and exit the routine (Abort) is left to the user.

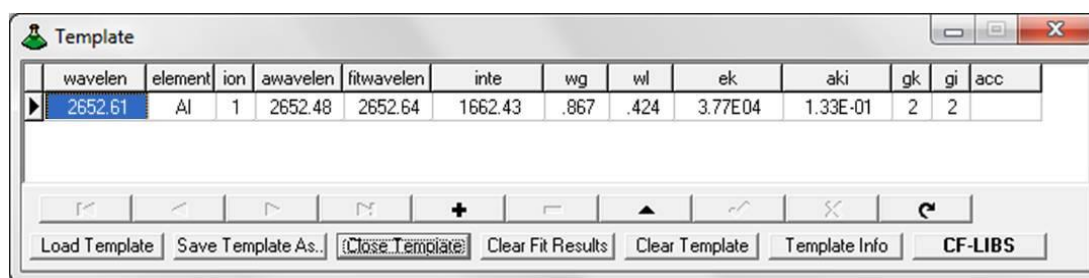


Beside the χ^2 parameter, the user is provided with a further tool to assess the quality of the fit: by pressing the Show Residuals button, the difference between experimental and analytical (resulting from the fit) profiles is displayed in a separate window.

Let's now consider the result of the fit satisfactory and accept it by the Stop button.

Only after this explicit acceptance of the results the fitting parameters are copied to the template.

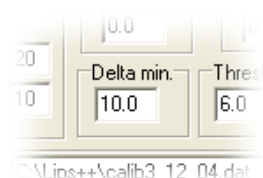
The parameters stored are: central wavelength of the analytical function (in the column fitwavelen), integral intensity of the analytical function calculated between the limits of the spectral interval currently displayed (in the column inte), Gaussian width and Lorentzian width (in the columns wg and wl, respectively).



At this step, the fields corresponding to the line from neutral aluminum at 2652.48 Angstrom are all completed. The information necessary for the quantitative analysis has been gathered. Of course, the same must be carried out for as many other lines in the spectrum as possible.

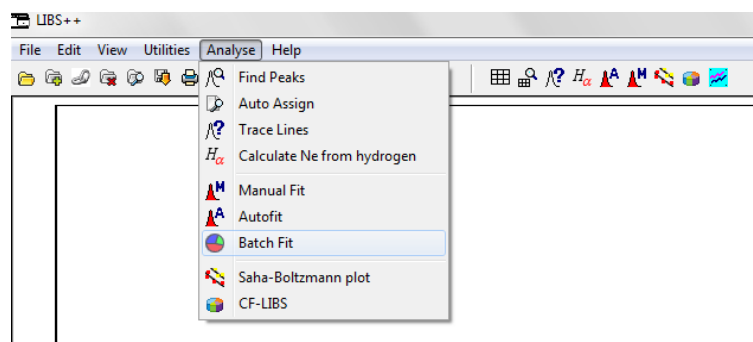
Let's pass now to the Autofit mode. By choosing this mode, the user delegates the management of the spectral intervals to be fitted to the software.

In fact, when pressing the Autofit icon , the spectral range displayed in the main window is subdivided in smaller intervals containing isolated marked lines or small groups of marked lines. The width of the sub-interval is determined case by case by clustering the marked spectral lines closer to each other than the distance Delta min. specified in the Options window. The algorithm which determines the sub-intervals is not sensitive to the presence of unmarked spectral lines.



To use the AutoFit mode, set first the largest view of the spectrum, to include all the marked lines in the main window. Then, click on the icon : the AutoFit algorithm will automatically perform the integrals of all the marked peaks and the results will be written in the template.

In the case a single template can be applied to many spectra, the AutoFit procedure can be replaced by the Batch Fit command. In order to use this option, first the template must be opened. Then, click on the Batch Fit button as shown in the figure:



Here, select the spectra to be analyzed and open them. The program will automatically perform the integrals for all the peaks in the templates and for all the spectra selected. At the end, a "Batch Fitting" window appears, in which the results of the integrations are shown; these results can be eventually saved by clicking on the dedicated button.

Anyway, the user should be aware that this procedure is not error-proof, since the fitting of the single peaks are shown very briefly on the screen and that does not allow to judge the goodness of the single fits. Therefore, this option should be only used for "trusted" spectra and templates, i.e., when one is absolutely sure that the fit of every peak will be done properly.

3.7 Plasma parameters

The application of the CF-LIBS method to quantitative analysis requires the knowledge of several characteristic parameters of the plasma (to a first approximation, plasma temperature and electron density). In many cases, the values of these parameters can be derived from some features of the spectrum itself. Let's assume that as more lines as possible have been marked, identified and fitted to evaluate their integral intensity.

3.7.1 Electron density

The information included in a broad band spectrum offers several possibilities for the evaluation of the electron density. One among them consists in exploiting the Saha equation; however, due to the intrinsic criticality, this topic will be covered in a later section.

The most direct approach to plasma electron density evaluation is the measurement of the Stark broadening of the spectral lines. In fact, even if different broadening mechanisms may be effective in increasing the width of emission lines in plasma, in typical LIBS conditions the Stark effect is often the most prominent. Thus, to a first approximation, the observed line width can be considered as the convoluted effect of Stark broadening and instrumental broadening. If the second is known, the first can be derived, and the electron density can be evaluated in turn by applying the commonly used formula (see section....) that establishes a simple relationship between electron density and line width.

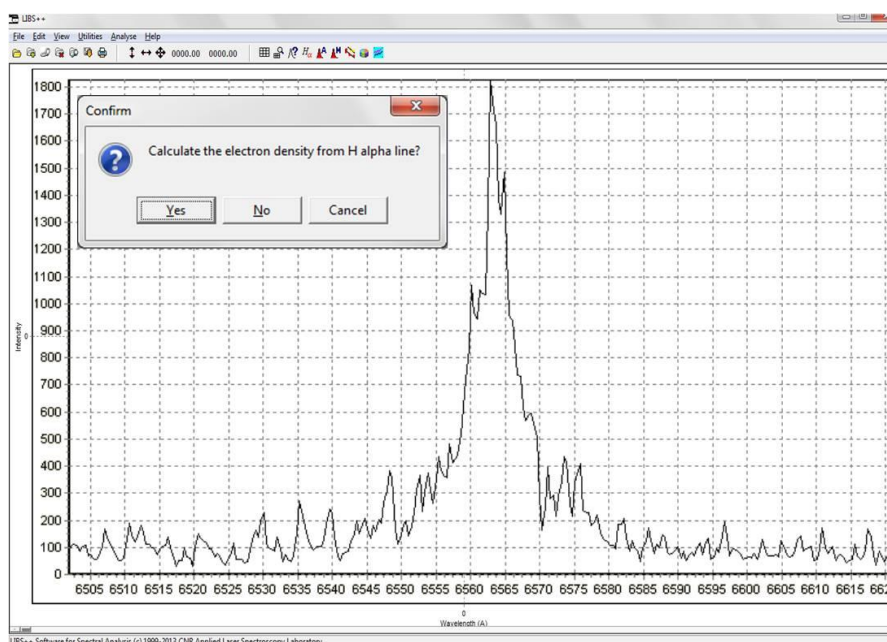
However, each emission line is affected by electron density to a different extent; this phenomenon is described by defining a Stark broadening coefficient: each line, then, is characterised by its own Stark coefficient. A Stark database is provided with LIBS++, containing the Stark broadening coefficients of several intense lines from some common elements. The evaluation of the electron density can be performed

according to this approach if at least one of the lines of the Stark database can be detected in the current spectrum.

However, if allowed by the spectral range included in the spectrum and by the environment of the measurement, the calculation of the electron density from the Stark broadening of the Hydrogen alpha line at 6563 Å is strongly recommended. This line is commonly present in the spectrum even if the hydrogen is not a component of the sample investigated, provided the measurement has been performed in air.

The H α line width ranges commonly between 5 and 30 Å, compared to about 1 Å of the other lines. This allows lower relative uncertainties on the results of the fit and deconvolution from the instrumental broadening.

The approach based on the H α line width is implemented as a separate tool, and can be accessed by the icon H_{α} in the command bar or by the menu command *Analyse/Calculate Ne from hydrogen*. First of all, a zoomed view of the spectral region including the hydrogen line wavelength is displayed in the main window, to allow the user checking about the presence of the emission.

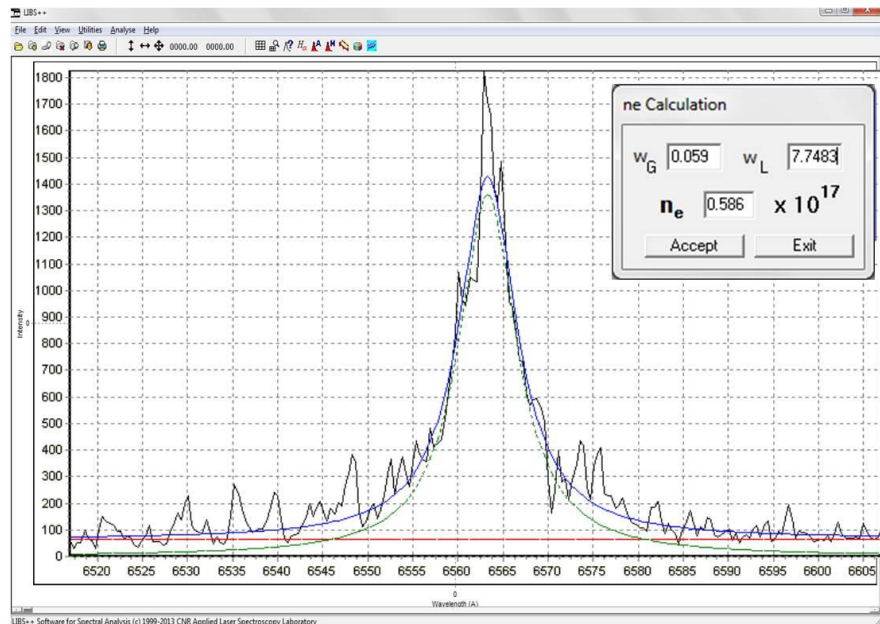


For example, in the spectrum chosen as the case study, the emission from the H α line is easily recognised. Since hydrogen is not a component of the aluminium alloy, the emitting atoms must have been provided by the ambient gas (in the specific case: air).

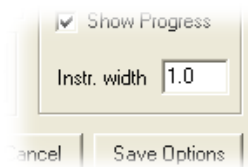
In a case like the one shown, the user can safely give the confirmation (Yes button). Then, the fitting of the hydrogen alpha line is started.

Similar to the general case of line fitting, a linear function (in red) is taken to represent the background, a Voigt function is used to fit the line and the total analytical profile is drawn in blue.

At the end of the fitting procedure, a report window (named “ne Calculation”) is shown, listing the line width contributions (Gaussian and Lorentzian) and the resulting electron density.



Before evaluating the electron density, the software must calculate the net line width by deconvolving the known instrumental broadening from the observed line width. This is done automatically, on the basis of the value stored in the Instrum.width box in the Options window. The correct use of this facility requires the user to type in this box the known instrumental contribution to the line width (expressed in Angstrom) at a wavelength of 5000 Angstrom. In the case shown in figure, the instrumental broadening at 5000 Å is 1Å. This means that a generic emission line centred at 5000 Å would be broader than 1 Å, since the instrumental systematic broadening would contribute for 1 Å width, and the Stark effect would add its positive contribute to the line width, not to speak about other possible effects. To evaluate the instrumental broadening contribution at different wavelength positions, the software automatically rescales linearly the known value. This is based on the assumption, valid for an echelle spectrometer, that:



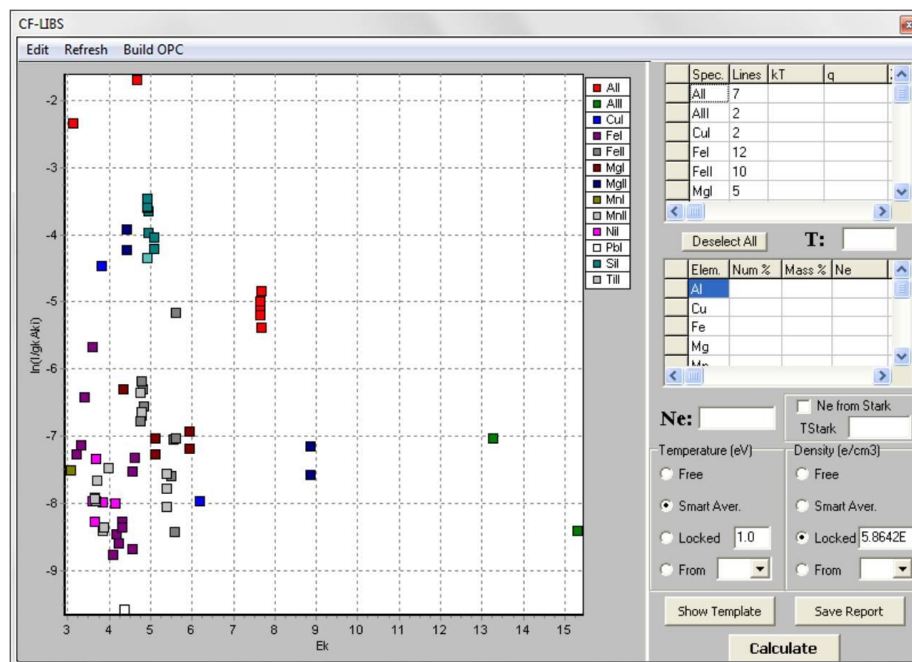
$$\left. \frac{\Delta\lambda}{\lambda} \right|_{instr} = const.$$

In the case of the hydrogen line at 6563 Angstrom, the instrumental broadening is determined by the software as about 1.3 Å.

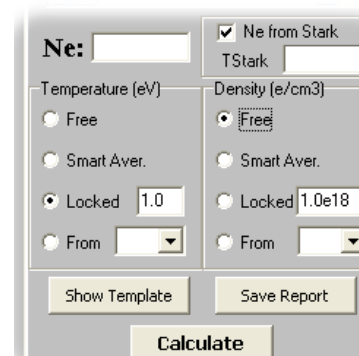
When the Accept button in the “ne Calculation” window is clicked, the ne value is stored for use in the following calculations.

Let’s switch now to the case that hydrogen emission line is not available. Other lines from heavier elements can be used to evaluate the Stark broadening and, in turn, the plasma electron density.

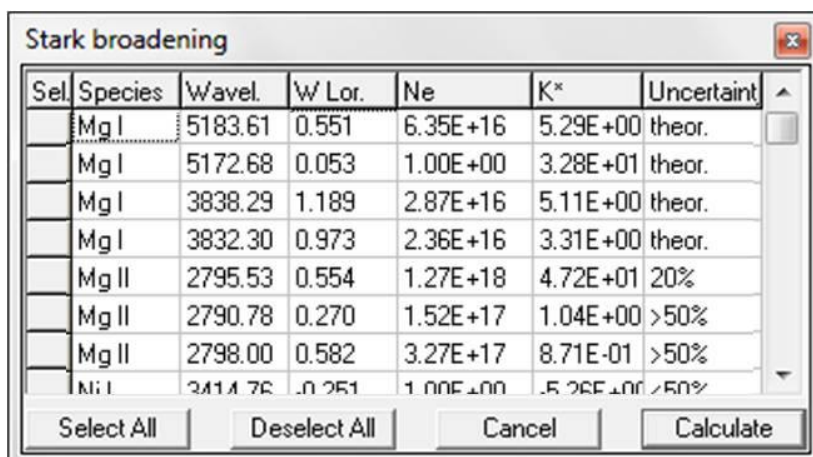
To carry out the electron density evaluation by this approach, open first the CF-LIBS window by the corresponding icon . The window will look like this:



However, at this time, only the box “Ne from Stark” should be considered. The other features will be described later. Since the Stark broadening is smoothly temperature-dependent, an initial guess of the plasma temperature must be assumed. The default value should be 1 eV. Let's lock the temperature and check the box “Ne from Stark”, then click on the Calculate button.



This image shows a close-up of the control panels from the CF-LIBS window. The 'Ne from Stark' checkbox is checked. The 'Temperature (eV)' is set to 'Locked' with a value of 1.0. The 'Density (e/cm³)' is set to 'Free'. The 'Calculate' button is visible at the bottom.



The Stark broadening window displays a table with the following data:

Sel	Species	Wavel.	W Lor.	Ne	K*	Uncertain
	Mg I	5183.61	0.551	6.35E+16	5.29E+00	theor.
	Mg I	5172.68	0.053	1.00E+00	3.28E+01	theor.
	Mg I	3838.29	1.189	2.87E+16	5.11E+00	theor.
	Mg I	3832.30	0.973	2.36E+16	3.31E+00	theor.
	Mg II	2795.53	0.554	1.27E+18	4.72E+01	20%
	Mg II	2790.78	0.270	1.52E+17	1.04E+00	>50%
	Mg II	2798.00	0.582	3.27E+17	8.71E-01	>50%
	Ni I	3414.76	0.251	1.00E+00	5.28E+00	>50%

At the bottom of the window are buttons for 'Select All', 'Deselect All', 'Cancel', and 'Calculate'.

The result will be shown in a new window named “Stark broadening”.

The window will provide, in separate rows, the ne estimates obtained by each line of the template which is also listed in the Stark database.

For each line in the list, the columns provide the following information:

Species – emitting element and ionization stage of the line;

Wavel. – central wavelength of the line in Angstrom;

W Lor. - net line width obtained after deconvolving instrumental broadening as described;

Ne - is the resulting electron density in cm^{-3} ;

*K** – this parameter is the Equivalent Optical Thickness (EOT) of the line, calculated as described in section.... This parameter is provided here to help the user in discriminating the most reliable lines among those included in the list. Ne estimations given by low EOT lines should be preferred because less affected by self absorption artefacts.

Uncertain. - The last column in the window quotes the uncertainty affecting the Stark coefficient for the corresponding line.

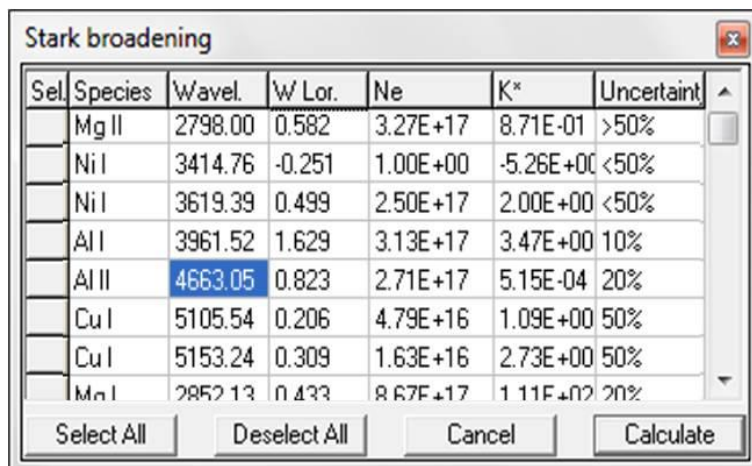
Several possibilities are now open to the user.

The average of the electron density values obtained from the different lines can be calculated: this is carried out by default when the user press Calculate in the Stark broadening window.

Alternatively, the user can unselect the lines considered less reliable just by clicking on the corresponding row in the Stark broadening window: an x sign will be added on the left of the row. After defining the reduced set of “reliable” lines, the command “Calculate” will provide the average of the corresponding ne values and will store the result for use in the following calculations.

The buttons Select All and Deselect All help in the batch operations, while single lines can be selected or unselected by repeated clicking of the left mouse button.

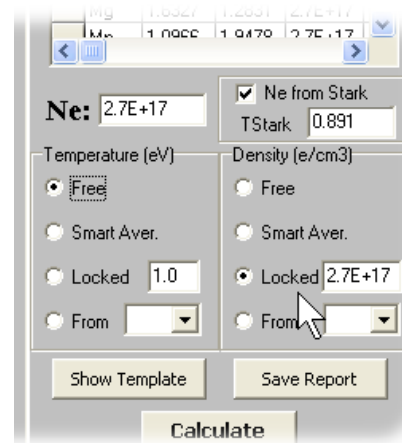
Coming back to the test spectrum, by scrolling down the table a line with very low EOT is found. It is the ionised aluminium line at 4663 Å: let's assume that this is the most reliable line among the available ones.



Sel.	Species	Wavel.	W Lor.	Ne	K*	Uncertain
☐	Mg II	2798.00	0.582	3.27E+17	8.71E-01	>50%
☐	Ni I	3414.76	-0.251	1.00E+00	-5.26E+00	<50%
☐	Ni I	3619.39	0.499	2.50E+17	2.00E+00	<50%
☐	Al I	3961.52	1.629	3.13E+17	3.47E+00	10%
☐	Al II	4663.05	0.823	2.71E+17	5.15E-04	20%
☐	Cu I	5105.54	0.206	4.79E+16	1.09E+00	50%
☐	Cu I	5153.24	0.309	1.63E+16	2.73E+00	50%
☐	Mn I	2852.13	0.433	8.67E+17	1.11E+02	20%

Buttons: Select All, Deselect All, Cancel, Calculate

After deselecting all the lines and selecting back only the 4663 Å line, the click on the Calculate button makes the software to store the corresponding n_e value for next calculations. This is confirmed by a change in the CF-LIBS window, where, now, the n_e box is locked to the chosen value.



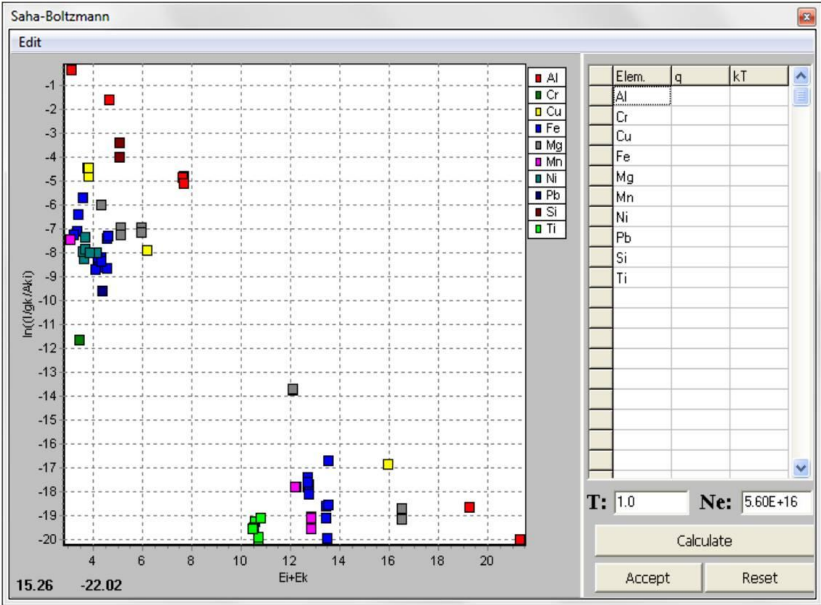
3.7.2 Temperature

After the identification of the spectral lines and the calculation of the integrals is complete, the plasma temperature can be calculated from the Boltzmann plot. In fact, in the hypothesis of Local Thermodynamic Equilibrium, the population of the excited levels is described by the Boltzmann distribution. Since the intensity of the emission lines is proportional to the population of the upper level, the distribution of the population and therefore the temperature can be obtained by the analysis of the spectrum itself.

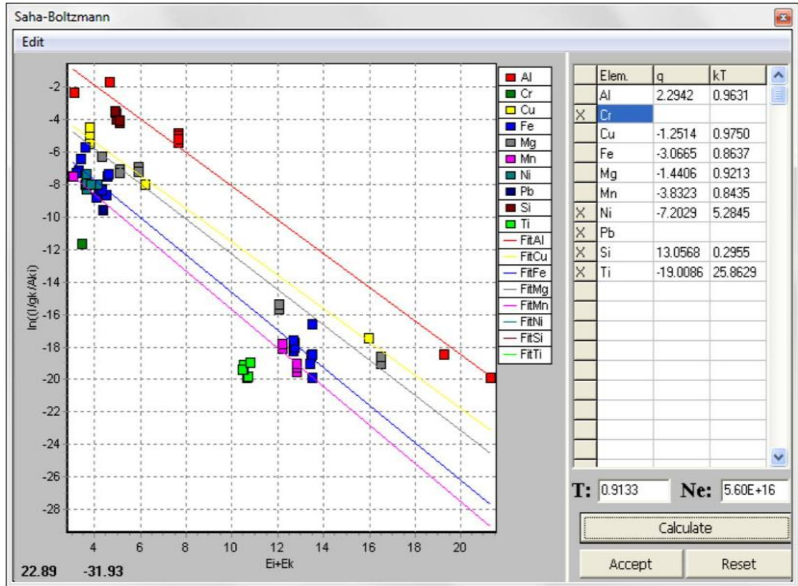
However, depending on the quality of the spectrum, the accuracy of the value determined by the Boltzmann plot may be unsatisfactory. Some improvement can be obtained by using a generalization of the Boltzmann plot, namely the Saha-Boltzmann plot, where lines from the neutral and the ionized species of the same elements, properly normalized, are combined to provide a wider dataset.

The Saha-Boltzmann plot of the emission lines detected in the spectrum can be shown by using the icon  or the menu command Analyse/Saha-Boltzmann plot. A new window is open, including the plot and the table where the resulting values of the temperature will be reported after the calculation is performed.

In the plot, each data point corresponds to one spectral line of the template (only the lines with known spectral parameters, however, can be displayed in the Saha-Boltzmann plot and used for quantitative analysis). The points corresponding to the same element are in the same colour. These datasets, including lines from the neutral and ionised stage, are spread across wider energy intervals compared to the ones in the normal Boltzmann plot. In general, this allows more accurate results.



However, a pre-requisite for the use of this approach is the knowledge of the electron density. The ne value should be set in the proper box, before starting the linear regression by means of the button “Calculate”. If the electron density has been already evaluated according to instructions in section 3.7.1, no further action is required and the calculation can be started directly. As a result, different T values will be obtained from the different element available, as shown in the right side table here below.



The table lists the element datasets available for the calculation and, as a result of the linear regression, the value of the intercept (q) and the reciprocal of the slope (kT), which is a measure of the temperature in units of eV.

The T box shows the average of the temperature (more precisely, the average of the kT values) obtained including the items marked by the x sign.

Let's consider our case study. The software automatically deselected several elements (they can be recognised by the x sign): Cr and Pb, because only one line is present in the template (no linear regression can be performed); Ni, Si and Ti, because the results of the regression are out of the acceptance limits currently set in the Options window. By the way, the latter three elements had lines from only one ionization species in the plot.

From: 2000 To: 5000
 Temperature Range: 0.3 < kT < 5.0
 electron density Range: 1.00e+1 < Ne < 1.00e+1

The datasets of the remaining elements (Al, Cu, Fe, Mg and Mn) all include data from both neutral and ionised states: these elements should provide in principle accurate results. Indeed, the temperature values obtained are not far from each other. However, if needed, the user can exclude or include in the average each element independently, just by a click of the left mouse button on the corresponding row in the table.

After changing the list of the selected elements, the linear regressions are calculated as before, but only the wanted values are averaged to give the resulting temperature. For instance, after unselecting Cu, the average T value is 0.8965.

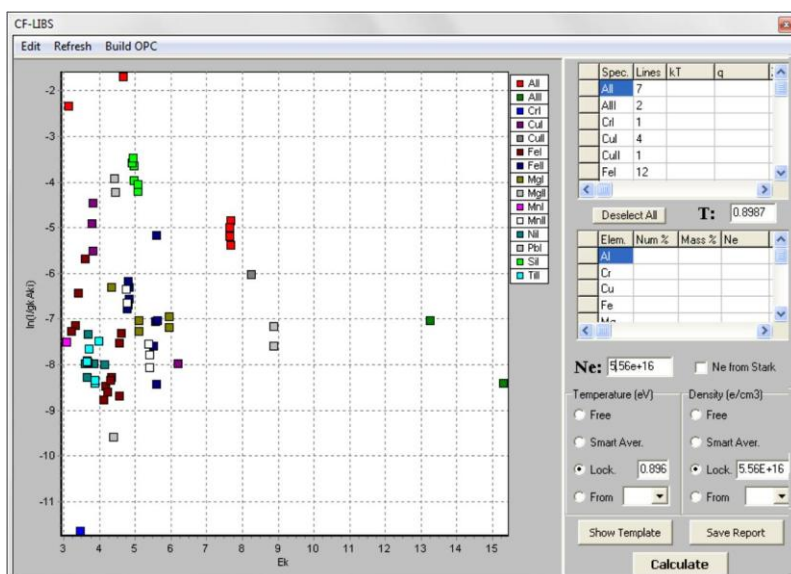
If accepted by clicking on the Accept button, the average value is stored and used in further calculations.

3.8 Quantitative analysis

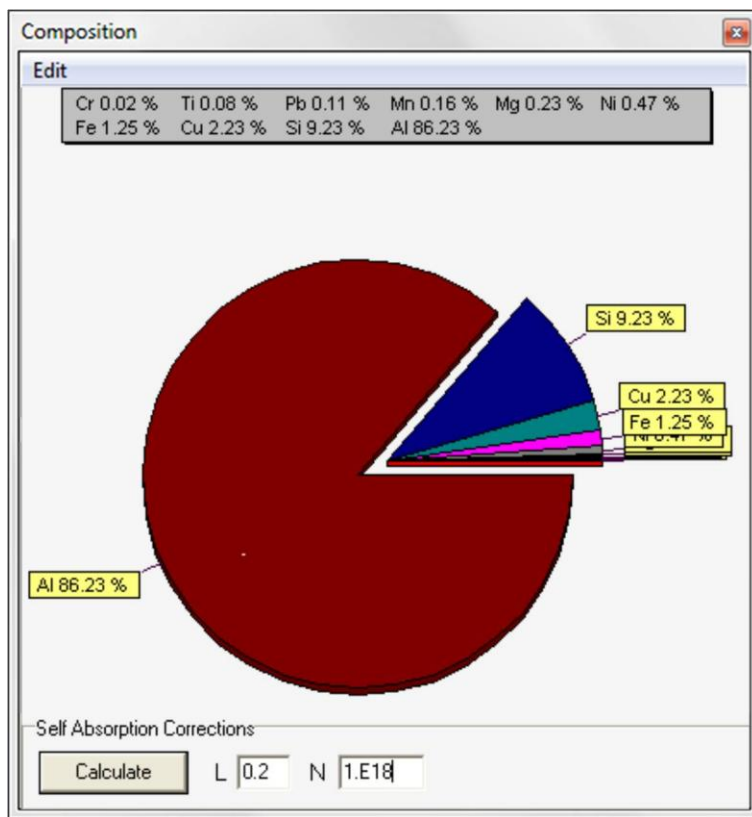
The quantitative analysis through the CF-LIBS method is activated by the icon  in the main window or by the command CF-LIBS in the template window. This will open the corresponding window, showing the Boltzmann plot resulting from the information stored in the template.

To proceed with the calculation, the plasma temperature and electron density must be available. Assuming that the steps described in sections 3.7.1 and 3.7.2 have been followed, the proper values are already set in the corresponding “Locked” boxes, which should appear checked.

The box “Ne from Stark” must be unchecked for the quantitative analysis.



The quantitative CF-LIBS calculation is activated by the command Calculate.



The result is provided graphically as a pie chart in a new window labeled "Composition" as well as numerically in the second table of the window CF-LIBS.

Deselect All				
Elem.	Num %	Mass %	Ne	
Al	88.6565	86.2291	5.56e+16	
Cr	0.0093	0.0175	5.56e+16	
Cu	0.9747	2.2325	5.56e+16	
Fe	0.6215	1.2511	5.56e+16	
Mg	0.2351	0.2251	5.56e+16	

The relative concentrations of the different elements are given (in alphabetic order): the sum of the concentration values of all the elements detected in the spectrum is set equal to 1. Both numeric and weight relative concentrations are given in the columns labelled Num% and Mass %, respectively.

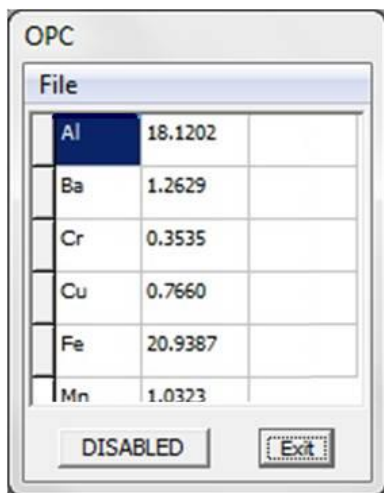
The values used to draw the pie chart are the relative concentrations in weight.

When the list of emitting elements included in the template is incomplete with respect to the real composition of the sample, it is still possible to use CF-LIBS in determining the relative concentrations of the detected elements with respect to each other.

The compositional result obtained so far may be satisfactory or not, depending on the features of the specific spectrum: in many practical cases, lines emitted by the major components are self-absorbed, leading to a biased evaluation of both temperature and concentration. In these cases the effect of self-absorption must be evaluated and possibly corrected.

3.8.1 Quantitative analysis with One Point Calibration (OPC)

The necessary condition to perform a CF-LIBS analysis is the knowledge of the response function of the spectrometer. In the case it is not known, quantitative analysis can be done as well, provided that the samples to be analyzed are similar and the composition of one of them is exactly known. The procedure is the following.



The "certified" sample is analyzed following the steps from paragraph 3.3 to 3.8, i.e., a normal CF-LIBS analysis is done. At the end, in the CF-LIBS window, click on the third button from the left in the toolbar ("Build OPC"). This will open a window where in the first column there are the elements, in the second column the concentrations just calculated with the CF-LIBS and in the last column the certified concentrations. Then click first on the "ENABLE" button, then on "CALCULATE" on the CF-LIBS main window and last on File -> Save STD in the OPC window.

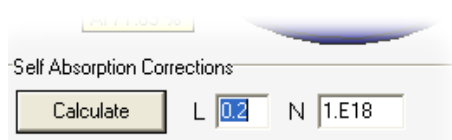
This will save a new response curve that will be used for all the other samples whose composition is similar to the certified one.

Therefore, in order to perform the quantitative analysis of the other samples, first replace the old response function in Edit -> Options with the new one (see sect. 3.1). Then repeat all the steps from sect. 3.3 to 3.8 for every sample.

IMPORTANT: It is mandatory to use for every sample the same peaks used for the reference one! A suggestion would be to use the same template, saving it with a different name every time the integrals for every sample have been calculated.

3.9 Evaluation of Self-absorption

After calculating the first estimate of the concentration by the CF-LIBS algorithm, as described in section 3.8, the user may want to evaluate whether self-absorption is affecting the intensity of the observed lines or not. This task is accomplished by the routine "self-absorption correction", executed by clicking on the Calculate button in the window "Composition".



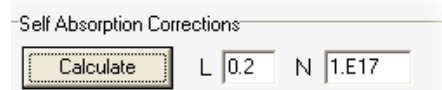
Beside the button, two boxes are available to input the values of the optical path length (L) and plasma number density (N). The optical path length must be expressed in cm; the plasma number density in cm^{-3} .

The optical path length depends on the size of the emitting plume at the time of the acquisition and on the geometric arrangement of the collection system. The size of the plume at a given time after the laser pulse depends in turn on the specific experimental conditions (buffer gas pressure, laser energy, acquisition

delay...). Whenever the possibility exists, it is advisable to measure the plume size in the specific experimental setup by gated imaging. The default value set in the L box (0.2 cm) corresponds to the plume radius in atmospheric pressure conditions, obtained after irradiation by pulse energy in the order of 100 mJ, and acquisition delay in the order of a couple of microseconds.

The plasma number density at the spectrum acquisition time can be measured as well by proper diagnostics. However, to a first approximation, an estimation of this quantity may be obtained by the spectrum itself.

As a very rough rule of thumb, the plasma number density can be approximated as the double of the electron density value. In the case of the test spectrum this is about $1 \cdot 10^{17}$.



After the execution of the Calculate command, a table is displayed in the window “Self Absorption Correction”, listing the self absorption coefficients evaluated for all the lines displayed in the Boltzmann plot. Lines are listed in rows, sorted by alphabetic order of the emitting element, while columns provide the coefficients and other complementary information.

The self absorption coefficient, reported in the third column (labelled “Corr.”), is defined as the ratio of the observed intensity over the expected intensity in case of optically thin plasma. When the coefficient is close to 1, the line is not self-absorbed; when the coefficient is $\ll 1$ the line is suspected to be strongly self absorbed.

Species	Wavel.	Corr.	Wl	Wl0
All	2652.48	0.9595	0.8495	0.8092
All	3050.07	0.9879	0.6385	0.6295
All	3054.68	0.9483	0.1427	0.1341
All	3057.14	0.8950	0.1766	0.1550
All	3064.29	0.9631	0.1828	0.1749
All	3066.14	0.9763	0.3273	0.3183
All	3961.52	0.3053	1.6294	0.4046
All	3586.00	0.9964	1.1834	1.1784
All	4663.05	0.9980	0.8227	0.8208
CrI	3578.69	0.9999	1.2898	1.2898
CuI	3247.54	0.8503	0.4559	0.3769
CuI	3273.95	0.8953	0.3298	0.2896
CuI	5105.54	0.9942	0.2061	0.2046
CuI	5153.24	0.9852	0.3094	0.3040
CuII	2247.00	0.8715	0.3214	0.2735

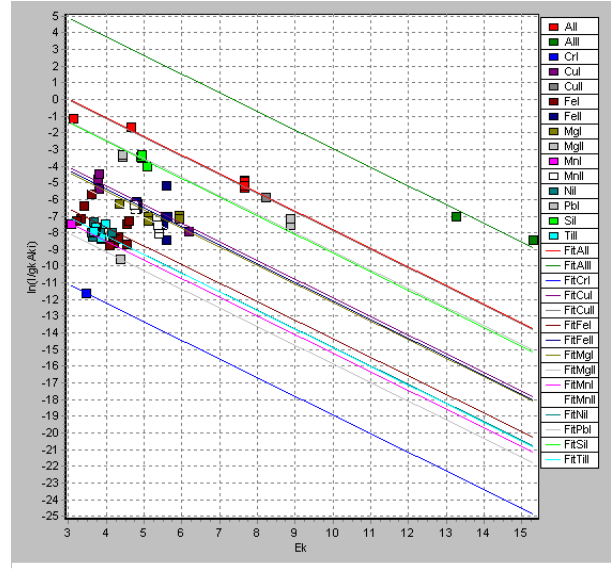
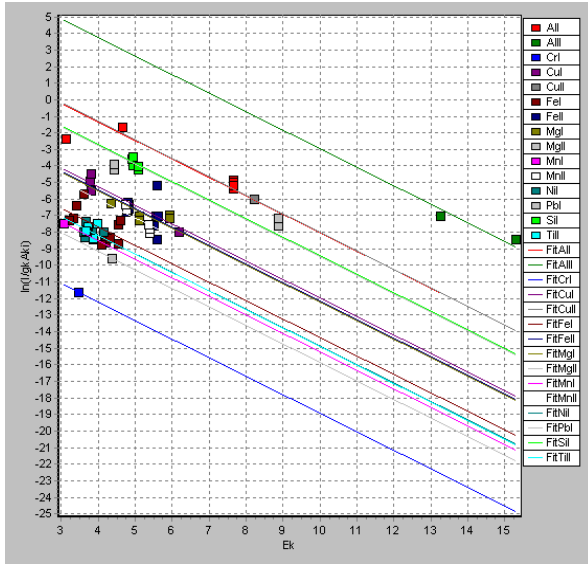
The column Wl provides the net width of the line, obtained by convolution of the two components in the analytical function (Gaussian and Lorentzian) and deconvolution of the instrumental with contribution. Opposite to this, which is the results of the measurement, the column Wl0 provides the net line width value expected in the absence of self-absorption. This value is calculated as described in section ...

As a consequence of a strong self absorption, the observed line intensity is lower than expected according to the population of the corresponding level. The contribution of that line to the Boltzmann plot is thus biased. The correction of this effect may improve the accuracy of the quantitative results.

The decision about the opportunity to apply the correction is however left to the user, after evaluating the values displayed in the third column of the table.

To apply the corrections, just press the CF-LIBS button in the Self absorption Corrections window; to ignore the correction and return to the CF-LIBS window, press "Cancel".

Looking into the table which shows the results obtained for the test spectrum, it is evident that at least several lines from aluminium are affected by self-absorption. Let's apply therefore the correction. When the "CF-LIBS" command in the Self Absorption correction window is given, the corrected line intensity values are calculated and used to adjust the position of the data points in the CF-LIBS window and in the Saha-Boltzmann plot. The comparison of the Boltzmann plot before and after applying the SAC is shown below.

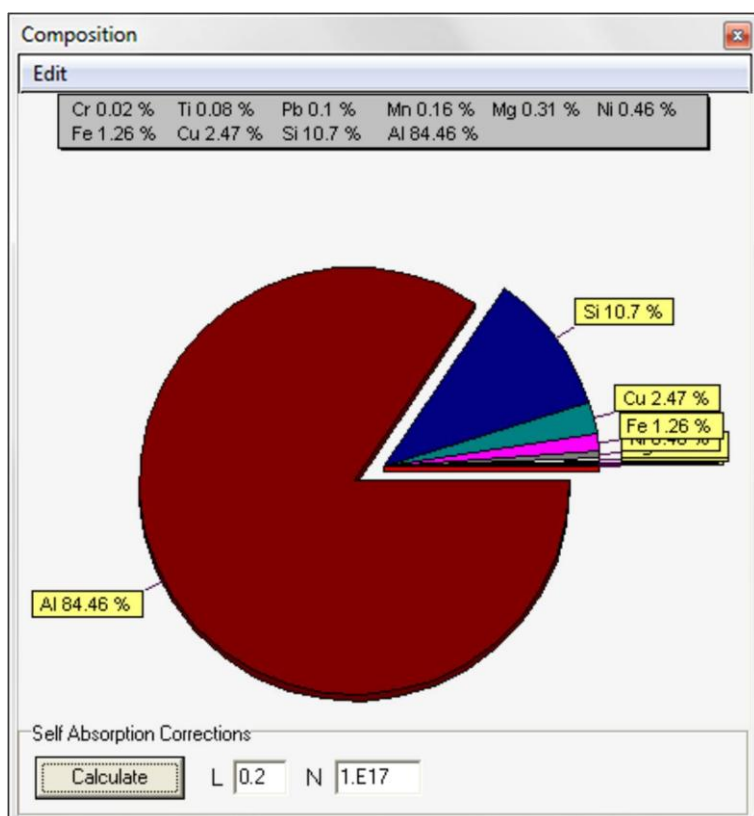


The most evident effect is the correction applied to the aluminium low energy line, which has been brought to a better alignment with respect to the others.

The corrected intensity values are not stored in the template. They remain available until the CF-LIBS window is open. When the CF-LIBS window is closed and open again, data for the plot will be retrieved from the template, and they will be always the uncorrected ones.

Intensity values are corrected also in the Saha-Boltzmann plot, so that a more reliable estimate of the temperature can be achieved. After running the Self Absorption correction, the plasma temperature can be calculated as described in section 3.7.2. In the test case the temperature was already calculated on the base of reliable data, so the SAC routine leads to a relative change in temperature lower than 0.4 %, which is negligible.

With the refined estimate of the temperature and the corrected intensity values, the composition can be evaluated by the same cf-librs algorithm as described in section 3.8.



The new composition result, obtained after SAC, is shown here.

A comparison of the results obtained by CF-LIBS before and after SA corrections to the certified values is provided in the table below. Concentration values are provided in weight %.

Some general comments follow.

The Self Absorption correction is effective in improving the accuracy of the concentration estimate for aluminium and magnesium, i.e. the two elements whose lines are in practice self-absorbed.

The concentration estimate of the most abundant elements (Al and Si) is accurate within a few percent. The other elements

are within 20-30% from the certified value. The certified composition includes also Sn and Zn which could not be detected in the studied spectrum.

Element	CF-LIBS	CF-LIBS & SAC	Certified
Al	86.23	84.46	84.42
Cr	0.02	0.02	0.032
Cu	2.23	2.47	2.93
Fe	1.25	1.26	0.9
Mg	0.23	0.31	0.329
Mn	0.16	0.16	0.19
Ni	0.47	0.46	0.356
Pb	0.11	0.1	0.059
Si	9.23	10.7	10.18
Ti	0.08	0.08	0.114
Total	100.01	100.02	99.51
			(+ Sn and Zn=100)

